

Motion in a Straight Line

Student Core • consolidated textbook

Concept → representation → equation → guided practice → independent transfer



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Reference Frame, Distance and Displacement

M-ST01A

See it -> name it -> use it

Learning target

Separate path length from net change of position.

CORE IDEA - keep this visible while learning

DISTANCE

 Σ path lengths

follows the actual route

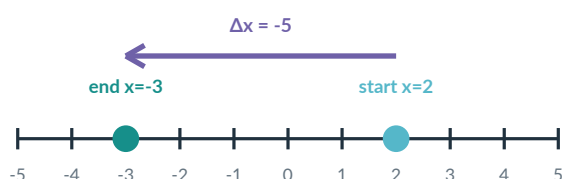
 ΔX
 $x_f - x_i$

endpoint change only

 $|\Delta X|$
 $\leq$ distance

always true

See the situation



Recognise it

- 1 start/end position is given
- 2 question asks distance vs displacement
- 3 motion reverses direction
- 4 closed trip / returns to start

FIRST MOVE

Choose an origin and + direction. Mark start, end, and actual path separately.

MEMORY ANCHOR

Distance follows the path. Displacement compares only start and end coordinates.

WHY IT WORKS

Distance = total path length. Displacement = final coordinate - initial coordinate. Always $|\text{displacement}| \leq \text{distance}$.

WORKED CONTRAST

Walk 4 m right, then 7 m left. Distance = 11 m. With right positive, displacement = -3 m.

COMMON TRAP

"One-dimensional motion means distance always equals displacement."

Do not set distance = $|\text{displacement}|$ if the path reverses or curves.

Guided practice

M-ST01A

Full support on the left; faded support on the right

USE THESE RELATIONS

DISTANCE

 Σ path lengths

follows the actual route

 Δx
 $x_f - x_i$

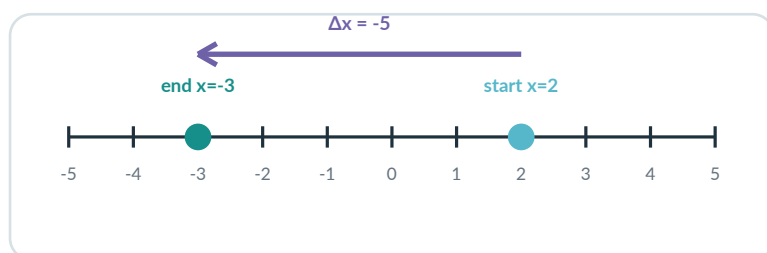
endpoint change only

 $|\Delta x|$
 $\leq$ distance

always true

Guided 1 - full support

A runner completes one lap and returns to the start. Which must be zero: distance or displacement?



MODEL -> CALCULATE -> CHECK

- distance = 6 + 2 + 5 = 13 m
- $\Delta x = +6 - 2 - 5 = -1$ m

H1: compare start/end. H2: closed path $\rightarrow$ endpoint change zero. H3: distance is path length.

EXPLAIN WHY THE RESULT MAKES SENSE

Guided 2 - now you do more

A student walks 6 m right, 2 m left, then 5 m left. Find distance and displacement.

DRAW THE PATH

DISTANCE

DISPLACEMENT

CHECK

IF STUCK

Separate the actual path from the start-to-finish change.

M-ST01A

Independent transfer

Choose the representation and model yourself

NO HINT INITIALLY

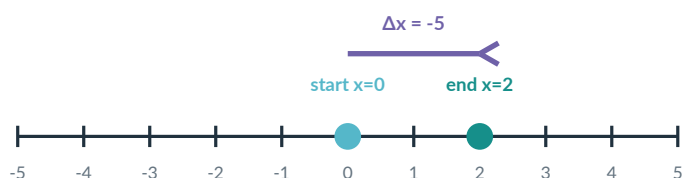
Challenge

A delivery rider travels 9 km and finishes 2 km east of the starting point. Which quantity tells you how much road was covered? Which tells you the net change of position?

YOUR FIRST MOVE

Draw a simple line or route. Mark start, finish, total path and net change separately.

1 Sketch the journey



2 Separate the quantities

ROAD COVERED (DISTANCE)

NET CHANGE (DISPLACEMENT)

UNITS

3 Explain

Why can these two quantities be different?

Circular Paths: Arc Distance vs Chord Displacement

M-ST01B

See it -> name it -> use it

Learning target

Separate arc distance from straight-line displacement.

CORE IDEA - keep this visible while learning

DISTANCE

$$\pi R$$

semicircle path

 $|\Delta R|$

$$2R$$

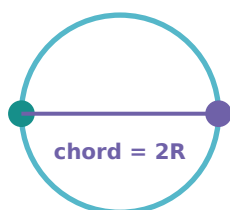
diameter chord

FULL TURN ΔR

$$0$$

endpoint returns

See the situation

semicircle arc = πR 

Recognise it

- 1 circle / semicircle / revolution language
- 2 path is curved but displacement asks endpoints
- 3 complete turns plus a final fraction

FIRST MOVE

Draw the circle, mark endpoints, trace the arc actually travelled, then draw the straight chord.

MEMORY ANCHOR

On a curved path, distance follows the arc while displacement is the straight chord joining endpoints.

WHY IT WORKS

For a half-circle: distance = πR , displacement magnitude = $2R$. Complete revolutions add path length but contribute zero net displacement.

WORKED CONTRAST

Travel a semicircle radius R . The journey length is πR , but endpoint separation is $2R$.

COMMON TRAP

"The object travelled along the displacement chord."

Do not use circumference or arc length when the question asks displacement.

Guided practice

M-ST01B

Full support on the left; faded support on the right

USE THESE RELATIONS

DISTANCE

$$\pi R$$

semicircle path

 $|\Delta R|$

$$2R$$

diameter chord

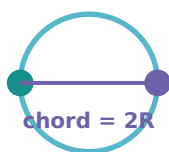
FULL TURN ΔR

$$0$$

endpoint returns

Guided 1 - full support

A body moves along a semicircle of radius R . Compare distance with displacement magnitude.

semicircle arc = πR 

MODEL -> CALCULATE -> CHECK

- distance: full circumference + quarter-arc
- displacement: straight chord between final endpoints

H1: path is an arc. H2: endpoints form a diameter. H3: write πR and $2R$.

EXPLAIN WHY THE RESULT MAKES SENSE

Guided 2 - now you do more

A body completes one full revolution and then a quarter revolution on a circle. What should you compute for distance and for displacement?

SKETCH ARC + ENDPOINTS

DISTANCE IDEA

DISPLACEMENT IDEA

CHECK

IF STUCK

Trace the arc first; then join the endpoints with one straight chord.

Independent transfer

M-ST01B

Choose the representation and model yourself

NO HINT INITIALLY

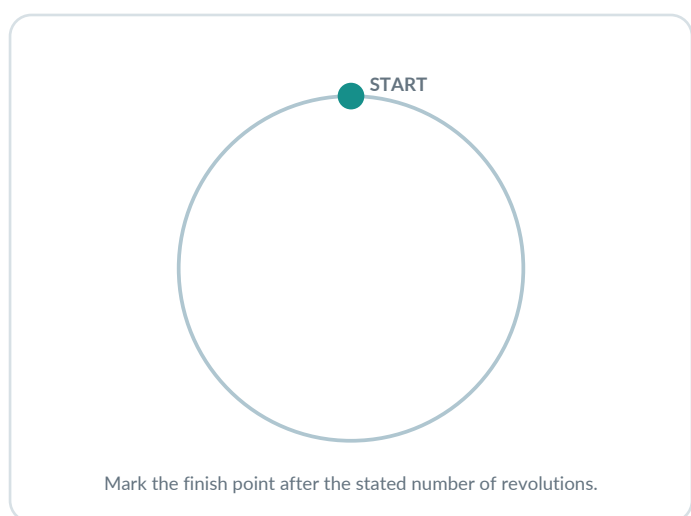
Challenge

An object moves around a circle for 2.5 revolutions. Which parts of the journey add to the distance travelled? Which parts determine the final displacement?

YOUR FIRST MOVE

Mark the start point, direction of travel and finish point after 2.5 turns. Do not write a formula yet.

1 Sketch and mark



2 Distance vs displacement

DISTANCE

Which parts of the path count?

DISPLACEMENT

Which two points matter?

3 Explain the role of complete revolutions

Use complete sentences: explain why full turns affect path length but not the final endpoint change.

Speed vs Velocity: Different Numerators

M-ST02A

See it -> name it -> use it

Learning target

Choose the correct numerator for average speed and average velocity.

CORE IDEA - keep this visible while learning

AVERAGE SPEED

total distance / total time

path numerator

AVERAGE VELOCITY

displacement / total time

endpoint numerator

See the situation



Recognise it

- 1 average speed vs average velocity
- 2 round trip
- 3 distance and displacement differ

FIRST MOVE

Write two numerators explicitly: total distance and displacement. They are not interchangeable.

MEMORY ANCHOR

Average speed uses total distance; average velocity uses displacement. A round trip can have nonzero average speed and zero average velocity.

WHY IT WORKS

average speed = total distance / total time; average velocity = displacement / total time.

WORKED CONTRAST

Go 30 m out and 30 m back in 10 s: average speed = 6 m/s; average velocity = 0.

COMMON TRAP

"Average speed and average velocity are the same formula."

Do not replace signed displacement with total path.

Guided practice

M-ST02A

Full support on the left; faded support on the right

USE THESE RELATIONS

AVERAGE SPEED

total distance / total time

path numerator

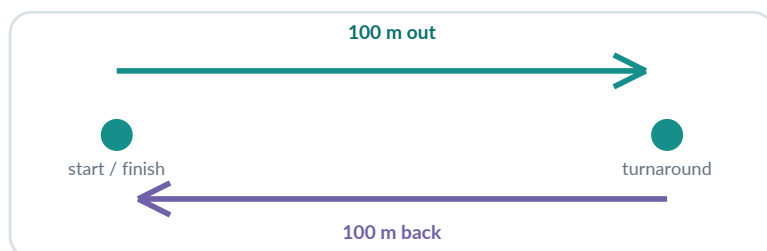
AVERAGE VELOCITY

displacement / total time

endpoint numerator

Guided 1 - full support

A cyclist travels 100 m out and 100 m back in 40 s. Find average speed and average velocity.



MODEL -> CALCULATE -> CHECK

- distance = $100 + 100 = 200$ m
- $\Delta x = 0$, $\Delta t = 40$ s
- average speed = $200 / 40 = 5$ m/s
- average velocity = $0 / 40 = 0$

H1: totals have different numerators. H2: displacement is zero. H3: divide each by 40 s.

EXPLAIN WHY THE RESULT MAKES SENSE

Guided 2 - now you do more

A motion has positive total distance but zero displacement. What must be true about average velocity?

TOTAL DISTANCE

DISPLACEMENT

AVERAGE VELOCITY

EXPLAIN

IF STUCK

Write the two numerators before you divide by time.

M-ST02A

Independent transfer

Choose the representation and model yourself

NO HINT INITIALLY

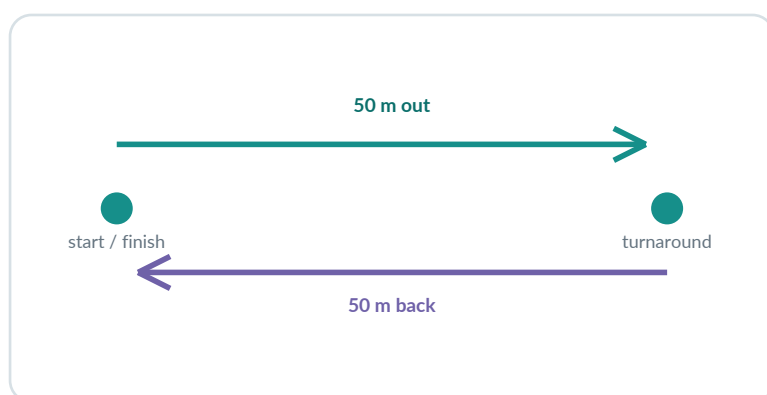
Challenge

Create your own round trip in which the average speed is nonzero but the average velocity is zero. Show the journey and explain why both statements can be true.

YOUR FIRST MOVE

Draw the journey first. Make sure the finish point is the same as the start point.

1 Build your round trip



2 Fill the two numerators

TOTAL DISTANCE

DISPLACEMENT

TOTAL TIME

3 Explain why the averages differ

Explain using the two numerators: total distance and displacement.

Equal Time vs Equal Distance Averages

M-ST02B

See it -> name it -> use it

Learning target

Recognise whether time or distance is being weighted equally.

CORE IDEA - keep this visible while learning

EQUAL TIMES v

$$(v_1 + v_2) / 2$$

equal time weights

EQUAL DISTANCES v

$$2v_1v_2 / (v_1 + v_2)$$

slower leg weighs more

GENERAL v

total distance / total time

fallback definition

See the situation

EQUAL TIME

v_1 for T

v_2 for T

same time weight → arithmetic mean

EQUAL DISTANCE

L at v_1

L at v_2

slower leg lasts longer

Recognise it

- 1 'equal times' or 'first T, next T'
- 2 'half the distance' / equal distances
- 3 same speeds but different weighting structure

FIRST MOVE

Mark what is equal: time or distance. Build total distance / total time before using any shortcut.

MEMORY ANCHOR

Equal times give equal time weights; equal distances give unequal times and a harmonic-type average.

WHY IT WORKS

Equal times: $(v_1 + v_2)/2$. Equal distances: $2v_1v_2/(v_1 + v_2)$. If neither condition holds, return to total distance / total time.

WORKED CONTRAST

10 and 15 m/s: equal times → 12.5 m/s; equal distances → 12 m/s.

COMMON TRAP

"Average of two speeds is always the arithmetic mean."

Do not use the equal-time or equal-distance shortcut unless that weighting condition is actually true.

Guided practice

M-ST02B

Full support on the left; faded support on the right

USE THESE RELATIONS

EQUAL TIMES $\bar{v}$

$$(v_1 + v_2) / 2$$

equal time weights

EQUAL DISTANCES $\bar{v}$

$$2v_1v_2 / (v_1 + v_2)$$

slower leg weighs more

GENERAL $\bar{v}$

$$\text{total distance} / \text{total time}$$

fallback definition

Guided 1 - full support

A cyclist moves at 8 m/s for 3 min and then 12 m/s for the next 3 min. Find average speed.

EQUAL TIME

 v_1 for T v_2 for Tequal time weights $\rightarrow$ arithmetic mean

MODEL $\rightarrow$ CALCULATE $\rightarrow$ CHECK

- same 3 min each $\rightarrow$ equal time weights
- $\bar{v} = (8 + 12)/2 = 10$ m/s

H1: times are equal. H2: equal time weights. H3: arithmetic mean.

EXPLAIN WHY THE RESULT MAKES SENSE

Guided 2 - now you do more

Half a journey is at 12 m/s and the other half at 18 m/s. Find average speed.

WHAT IS EQUAL?

CHOOSE RELATION

CALCULATE

CHECK

IF STUCK

Ask what is equal: time or distance?

M-ST02B

Independent transfer

Choose the representation and model yourself

NO HINT INITIALLY

Challenge

Two journeys use the same two speeds. In one, the times are equal. In the other, the distances are equal. Predict which average speed is smaller, then justify your prediction.

YOUR FIRST MOVE

Sketch two journeys with the same speeds but different weighting. Ask what is equal before calculating.

1 Compare the weighting

EQUAL TIME

 v_1 for T v_2 for Tsame time weight $\rightarrow$ arithmetic mean

EQUAL DISTANCE

L at v_1 L at v_2

slower leg lasts longer

2 Predict first

Which average is smaller?
Why should the slower speed matter more in the equal-distance case?

3 Justify with a relation

EQUAL-TIME AVERAGE

EQUAL-DISTANCE AVERAGE

CONCLUSION

Repeated Motion: Complete Cycles + Final Partial Cycle

M-ST02C

See it -> name it -> use it

Learning target

Use one complete cycle as a bookkeeping unit, then inspect the unfinished part.

CORE IDEA - keep this visible while learning

CYCLE PATH

$$F + B$$

distance within one cycle

NET PER CYCLE

$$F - B$$

forward positive

AFTER N FULL CYCLES

$$N(F - B)$$

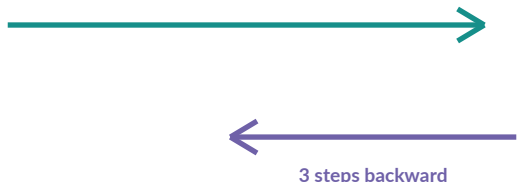
then inspect leftover motion

See the situation

ONE COMPLETE CYCLE

5 steps forward

net = +2 steps



Recognise it

- 1 repeated forward/back pattern
- 2 cycles or laps
- 3 target may be reached partway through final cycle

FIRST MOVE

Write one complete cycle: cycle time, cycle distance, net progress. Then handle the leftover motion.

MEMORY ANCHOR

Analyse one complete cycle, count full cycles, then inspect the unfinished final cycle.

WHY IT WORKS

One cycle creates a repeatable bookkeeping unit. Do not replace the final incomplete cycle by an average.

WORKED CONTRAST

5 forward, 3 back: one 8-step cycle gives +2 net steps. A target may be reached during the next forward run.

COMMON TRAP

"Use net progress per cycle and ignore where the target is reached inside the final cycle."

Do not assume the target is reached only at cycle boundaries.

Guided practice

M-ST02C

Full support on the left; faded support on the right

USE THESE RELATIONS

CYCLE PATH

$$F + B$$

distance within one cycle

NET PER CYCLE

$$F - B$$

forward positive

AFTER N FULL CYCLES

$$N(F - B)$$

then inspect leftover motion

Guided 1 - full support

A walker repeats 5 forward steps and 3 backward steps. After 3 complete cycles, what is the net progress?



MODEL -> CALCULATE -> CHECK

- one cycle: $+5 - 3 = +2$ steps
- 3 complete cycles: $3 \times 2 = +6$ steps

H1: build one cycle. H2: net +2 per cycle. H3: multiply by 3.

EXPLAIN WHY THE RESULT MAKES SENSE

Guided 2 - now you do more

The same walker wants to reach a point 9 steps ahead. Explain why counting complete cycles alone is not enough.

ONE-CYCLE NET

FULL-CYCLE PROGRESS

FINAL PARTIAL CYCLE

EXPLAIN

IF STUCK

A target can be reached during the next forward run, before another cycle ends.

WORK SPACE - REDRAW, CHECK SIGNS, OR FINISH YOUR EXPLANATION

M-ST02C

Independent transfer

Choose the representation and model yourself

NO HINT INITIALLY

Challenge

Create a forward/back repeated pattern in which a target is reached during the final incomplete cycle. Show exactly where the target is reached.

YOUR FIRST MOVE

Design one cycle first. Choose the forward move, backward move and target. Do not average away the final partial cycle.

1 Design one cycle

ONE CYCLE

forward = _____ steps



backward = _____ steps

2 Choose a target

TARGET POSITION

NET PER FULL CYCLE

NUMBER OF FULL CYCLES

3 Mark the final partial cycle

Show where the target is reached before the next full cycle finishes.

Acceleration as Change in Velocity

M-ST03A

See it -> name it -> use it

Learning target

Treat acceleration as signed change in velocity per unit time.

CORE IDEA - keep this visible while learning

VELOCITY CHANGE

$$\Delta v = v_f - v_i$$

subtract signed velocities

AVERAGE ACCELERATION

$$a_{\text{avg}} = \Delta v / \Delta t$$

change in velocity per time

UNIT

$$\text{m/s}^2$$

(m/s) per second

See the situation

t=0 s

t=1 s

t=2 s

t=3 s

4 m/s



6 m/s



8 m/s



10 m/s



equal +2 m/s change each second -> constant acceleration

Recognise it

- 1 velocity changes with time
- 2 same speed but direction changes
- 3 average acceleration asks for $\Delta v / \Delta t$

FIRST MOVE

Draw velocity arrows at start and finish before subtracting them.

MEMORY ANCHOR

Acceleration measures change in velocity per time; direction changes matter even when speed does not.

WHY IT WORKS

Average acceleration = (final velocity - initial velocity)/ Δt . Velocity is signed/directional.

WORKED CONTRAST

Velocity +12 to -4 m/s in 4 s: $\Delta v = -16$ m/s, so average acceleration = -4 m/s².

COMMON TRAP

"Acceleration only measures change of speed."

Do not subtract speed magnitudes when direction changes; subtract velocities.

Guided practice

M-ST03A

Full support on the left; faded support on the right

USE THESE RELATIONS

VELOCITY CHANGE

$$\Delta v = v_f - v_i$$

subtract signed velocities

AVERAGE ACCELERATION

$$a_{\text{avg}} = \Delta v / \Delta t$$

change in velocity per time

UNIT

$$\text{m/s}^2$$

(m/s) per second

Guided 1 - full support

A particle changes velocity from +12 to -4 m/s in 4 s. Find average acceleration.

INITIAL
+12 m/s

FINAL
-4 m/s



Subtract signed velocities, not speed magnitudes.

MODEL -> CALCULATE -> CHECK

- $\Delta v = -4 - (+12) = -16 \text{ m/s}$
- average acceleration = $-16 / 4 = -4 \text{ m/s}^2$

H1: use signed velocities. H2: $\Delta v = -4 - (+12)$. H3: divide by 4.

EXPLAIN WHY THE RESULT MAKES SENSE

Guided 2 - now you do more

A body moves at constant speed but reverses its velocity direction. Can average acceleration be nonzero? Explain.

DRAW INITIAL + FINAL V

WRITE SIGNED INITIAL/FINAL V

FIND Δv

CONCLUSION

IF STUCK

Speed can stay the same while velocity changes because direction changes.

WORK SPACE - REDRAW, CHECK SIGNS, OR FINISH YOUR EXPLANATION

M-ST03A

Independent transfer

Choose the representation and model yourself

NO HINT INITIALLY

Challenge

Give an example in which speed stays constant over an interval but average acceleration is not zero. Use velocity direction to explain why.

YOUR FIRST MOVE

Draw the initial and final velocity arrows before writing Δv . Keep the speed magnitudes equal if you want a direction-only change.

1 Draw two velocity states

INITIAL v	FINAL v
draw arrow + sign	draw arrow + sign

2 Compare speed and velocity

INITIAL SPEED

FINAL SPEED

INITIAL VELOCITY

FINAL VELOCITY

3 Explain the acceleration

Use signed Δv or a clear direction argument. Explain why unchanged speed does not force zero acceleration.

M-ST03B

Sign logic: negative acceleration can speed up

Direction first; signs second

BIG IDEA

Signs describe direction relative to the chosen axis. Speed increases when velocity and acceleration point the same way.

CASE A: v right, a left

velocity
acceleration



CASE B: v left, a left

velocity
acceleration



FIRST MOVE

Draw separate arrows for v and a . Compare directions, not just plus/minus symbols.

M-ST03B

Misconception clinic

Why negative a does not automatically mean slowing

COMMON WRONG IDEA

"Negative acceleration always means slowing down."

Why it feels plausible: many braking examples choose the direction of motion as positive, so braking happens to produce a negative acceleration.

CORRECT MODEL

Away from $v = 0$ in 1D:

- same direction for v and $a \rightarrow$ speed increases
- opposite directions $\rightarrow$ speed decreases

At $v = 0$, inspect what happens immediately after.

WORKED CONTRAST

EXAMPLE

$v = -6 \text{ m/s}$ and $a = -2 \text{ m/s}^2$. Both point left, so speed increases even though a is negative.

QUICK SIGN MAP

$v \rightarrow, a \rightarrow$

speed up

$v \rightarrow, a \leftarrow$

slow down

$v \leftarrow, a \leftarrow$

speed up

$v \leftarrow, a \rightarrow$

slow down

M-ST03B

Guided practice → independent transfer

Support fades from arrows to your own example

Guided 1 — full support

A body moves left with $v = -8 \text{ m/s}$ and $a = -2 \text{ m/s}^2$. Is its speed increasing or decreasing?

$v = -8 \text{ m/s}$
 $a = -2 \text{ m/s}^2$



H1: same sign. H2: arrows point the same way. H3: speed increases.

Guided 2 — faded

A body has $v = +10 \text{ m/s}$ and $a = -3 \text{ m/s}^2$. What happens to speed before it reaches $v = 0$?

DRAW V AND A ARROWS

SPEED CHANGE + REASON

INDEPENDENT TRANSFER — NO HINT INITIALLY

Give one example with negative acceleration that speeds up, and one that slows down. Draw the representation yourself before writing a relation.

EXAMPLE THAT SPEEDS UP

EXAMPLE THAT SLOWS DOWN

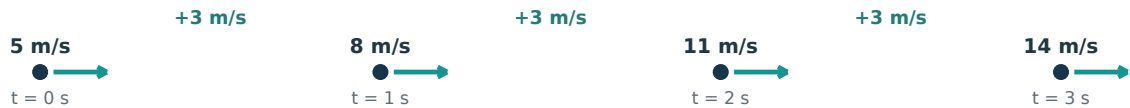
M-ST04A

Constant acceleration

Equal Δv during equal Δt is the visual signature

A SCOOTER LEAVING A TRAFFIC LIGHT

Velocity readings are 5, 8, 11 and 14 m/s at one-second intervals.



$$a = \Delta v / \Delta t$$

same velocity change in equal time intervals

WHAT DO YOU NOTICE?

Each second adds the same 3 m/s . The velocity is changing in a regular way. Constant acceleration means velocity changes by the same amount in equal time intervals.

PHYSICS MODEL, NOT A PROMISE

A train pulling away, a bicycle speeding up smoothly, or a car braking steadily can often be approximated by constant acceleration over a short interval. A real vehicle does not keep exactly the same acceleration forever.

CLOSE CONTRAST: ACCELERATION IS CHANGING

Readings 5, 8, 13 and 20 m/s give gains of $+3$, $+5$ and $+7 \text{ m/s}$. Equal time intervals do not give equal velocity changes, so one constant-acceleration value cannot describe the whole interval.

M-ST04A

Measuring acceleration

Use signed velocity change

FROM THE MOTION STORY TO A NUMBER

If velocity changes from +4 m/s to +10 m/s in 3 s, then $\Delta v = +6$ m/s.

$$a = (v - u) / t$$

$$a = (10 - 4) / 3 = +2 \text{ m/s}^2$$

READ THE UNIT

2 m/s² means velocity changes by 2 m/s during each second. The sign tells the direction of acceleration relative to the chosen positive axis.

DO NOT SUBTRACT SPEEDS

Suppose velocity changes from +12 m/s to -4 m/s in 4 s.

$\Delta v = \text{final} - \text{initial} = -4 - (+12) = -16$ m/s, not 8 m/s.

$a = -16 / 4 = -4$ m/s²

CONCEPT CHECK

The negative sign does not yet tell you whether the object is speeding up or slowing down. Compare the direction of v and a .

M-ST04A

The same motion in four languages

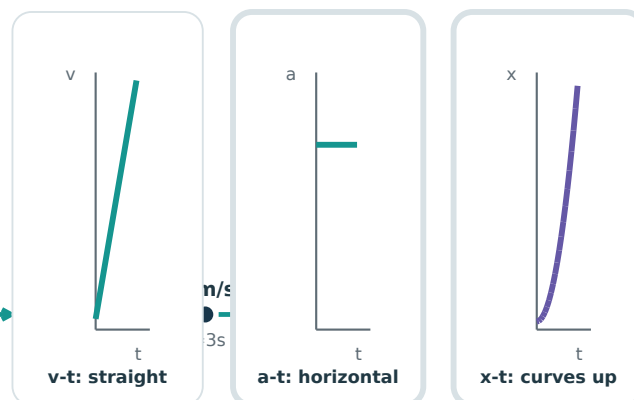
Words ↔ snapshots ↔ graphs

WORDS

Starts from rest and speeds up uniformly in the positive direction.



TRANSLATE INTO GRAPHS



MEMORY LINK

Uniform acceleration → equal Δv in equal Δt → straight-line v-t graph.

WHY X-T CURVES

As the object gets faster, it covers more displacement during each later second. So the position-time graph becomes progressively steeper.

M-ST04A

Build $v = u + at$

Reconstruct the relation instead of memorising it

SAY IT IN ORDINARY LANGUAGE

Begin with velocity u . Acceleration a is the velocity change per second. If it acts for t seconds, the accumulated velocity change is at .

STORY

final velocity = starting velocity + velocity gained

$$v = u + at$$

QUICK CHECKS

If $a = 0$, then $v = u$. If $t = 0$, then $v = u$. Units: $(\text{m/s}^2)(\text{s}) = \text{m/s}$, so at is a velocity change.

REBUILD IT FROM THE DEFINITION

$$1 \quad a = (v - u) / t$$

$$2 \quad at = v - u$$

$$3 \quad v = u + at$$

WHY THIS MATTERS

If the equation is forgotten, you can rebuild it. That is stronger than recalling a label such as “first equation of motion.”

M-ST04A

Worked example: a cyclist speeds up

Start + change = finish

SEE THE PHYSICAL SITUATION

A cyclist is already moving at 6 m/s. For the next 4 s, the cyclist accelerates uniformly at +2 m/s². Find the final velocity.

REASONING STORY

Start: 6 m/s

Change: 2 m/s each second $\times$ 4 s = 8 m/s

Finish: 6 + 8 = 14 m/s

UVATS

u	6 m/s
v	?
a	+2 m/s ²
t	4 s
s	not needed

$$v = u + at$$

$$v = 6 + (2)(4) = 14 \text{ m/s}$$

PHYSICAL CHECK

Velocity and acceleration point the same way, so the speed should increase. 14 m/s is consistent with the motion story.

M-ST04A

Misconception clinic

The sign of a gives direction, not “speeding/slowing” by itself

TEMPTING WRONG IDEA

“Acceleration is negative, so the object must be slowing down.”

Many braking examples choose the direction of motion as positive. The learner can accidentally attach “negative” to “slowing.”

WHAT THE SIGN ACTUALLY MEANS

A negative acceleration only means the acceleration arrow points in the negative direction of the chosen axis.

FAMILIAR BRAKING CASE

velocity
acceleration



arrows oppose → speed decreases

COUNTEREXAMPLE

velocity
acceleration



arrows agree → speed increases

CORRECT MODEL + RETRY

Away from the instant $v = 0$: same sign for v and a → speed increases; opposite signs → speed decreases.

Retry: a cart has $v = -6 \text{ m/s}$ and $a = -2 \text{ m/s}^2$. Predict the speed change before calculating.

M-ST04A

Application: how long until it stops?

Translate the story before choosing a relation

A BICYCLE BRAKES STEADILY

A bicycle is moving at 20 m/s. The rider brakes so that acceleration is a constant 5 m/s² opposite the motion. How long until the bicycle stops?

HIDDEN WORD

Stops → $v = 0$. Choosing motion to the right as positive makes the braking acceleration negative.

u

20

v

0

a

-5

t

?

s

not needed

$$0 = 20 - 5t \rightarrow t = 4 \text{ s}$$

$v = u + at$ connects exactly u, v, a and t; displacement is not needed.

WHEN NOT TO USE ONE CONSTANT-A EQUATION

DRIVER REACTS, THEN BRAKES

A driver sees an obstacle, continues at nearly constant speed for 0.8 s while reacting, and only then brakes with approximately constant acceleration. One constant-acceleration interval does not describe the entire reaction + braking journey. Split at the stage boundary and apply the correct model to each stage.

M-ST04A

A repeatable first-step routine

Make the relation obvious before arithmetic

1

SEE + FRAME

Describe the motion; choose a positive direction.

2

REPRESENT

Draw the smallest useful motion strip or arrows.

3

EXTRACT

Write u , v , a , t , s and translate hidden words.

4

MODEL

Check acceleration is constant for this interval.

5

CHOOSE + CHECK

Circle the target; avoid unnecessary variables; solve and check.

PHRASE DECODER

starts from rest

$$u = 0$$

stops / comes to rest

$$v = 0$$

uniform acceleration

$$a = \text{constant}$$

slowing down

compare v and a directions

distance irrelevant

$$v = u + at \text{ may be efficient}$$

MEMORY QUESTION

Do not ask “Which formula do I remember?” Ask: What is happening? What is known? What is asked? Which variable can I avoid?

M-ST04A

Guided 1 — full scaffold

Reveal hints only when needed

QUESTION

A scooter moves at 18 m/s and slows uniformly at 3 m/s². How long until it stops?

$v = u + at$

operative relation is visible on full-support guided pages

FILL THE STRIP

u	18 m/s
v	0 m/s ← stops
a	-3 m/s ²
t	?
s	not needed

BEFORE HINTS

Choose the positive direction and fill as much of the strip as you can.

HINT LADDER

H1 — NOTICE: “stops” gives $v = 0$.
H2 — MODEL: $u = 18, v = 0, a = -3, t = ?$.
H3 — START: $0 = 18 - 3t$. Finish the arithmetic yourself.

FINISH THE ARITHMETIC

CHECK SIGN, UNIT AND REASONABLENESS

M-ST04A

Guided 2 — faded support

You build more of the representation

QUESTION

A cart is moving left at 12 m/s. Its constant acceleration is 4 m/s² to the right. After how long will its velocity become zero?

YOUR JOB

Draw the velocity and acceleration arrows. Choose right as positive. Then write signed u , v , a and the target.

ARROWS + POSITIVE DIRECTION

FIRST RELATION

U, V, A, T

SOLVE + CHECK

HINTS — USE ONLY IF STUCK

- H1 — Leftward velocity is negative if right is positive.
- H2 — $u = -12\text{ m/s}$, $v = 0$, $a = +4\text{ m/s}^2$.
- H3 — Write $0 = -12 + 4t$.

M-ST04A

Independent transfer

Predict the speed change before calculating

CHALLENGE

A trolley has velocity -5 m/s and constant acceleration -1.5 m/s^2 . Predict whether its speed initially increases or decreases. Then find its velocity after 4 s.

FIRST MOVE

No diagram is provided. Draw your own velocity and acceleration arrows. Decide the speed change before calculating.

DRAW V AND A ARROWS

PREDICTION: SPEED INCREASES OR DECREASES? WHY?

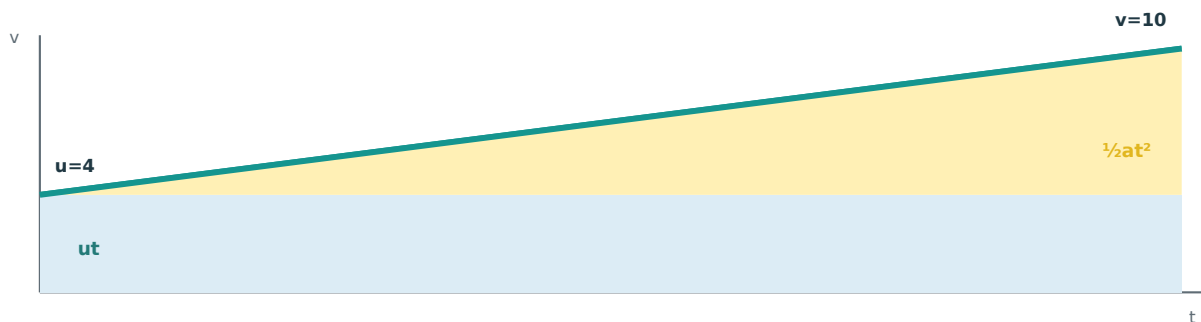
CALCULATION: FIND VELOCITY AFTER 4 S

PHYSICAL CHECK

M-ST04B

Displacement under constant acceleration

See displacement as area under v - t , then build $s = ut + \frac{1}{2}at^2$.



MEMORY ANCHOR

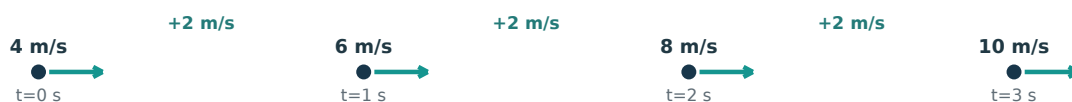
Displacement = area under the velocity-time graph. The area splits into starting-motion area + acceleration-added area.

WHAT WE WILL EXPLAIN

Why the area splits into ut and $\frac{1}{2}at^2$, why $\frac{1}{2}$ appears, and why time is squared.

REAL-LIFE STORY: THE CYCLIST COVERS MORE EACH SECOND

A cyclist starts at 4 m/s and accelerates uniformly at 2 m/s^2 . The velocity at successive seconds is 4, 6, 8, 10 m/s.



TWO CONTRIBUTIONS

If velocity stayed at 4 m/s for 3 s, displacement would be 12 m. Acceleration adds extra velocity continuously, so there is additional displacement. 1) displacement from the starting velocity u ; 2) extra displacement created by acceleration.

M-ST04B

Displacement is area under v-t

The graph stores accumulation

WHY AREA HAS THE RIGHT UNITS

Velocity has units m/s and time has units s. Multiplying them gives metres, the unit of displacement.

$$(\text{m/s}) \times \text{s} = \text{m}$$

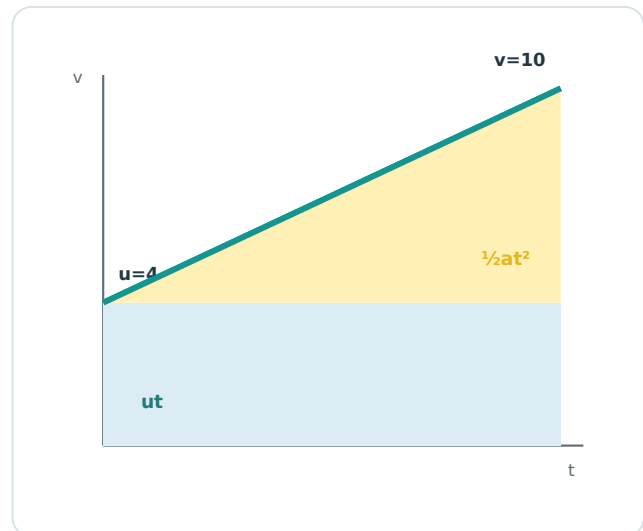
GRAPH HELPER

On a v-t graph, each thin vertical strip is approximately velocity $\times$ a tiny time interval. Adding all strips gives total displacement.

THINGS TO KNOW

Signed area gives displacement. If velocity goes below zero, that portion contributes negative displacement. Total distance is a later graph topic.

FOR CONSTANT ACCELERATION, THE GRAPH IS SIMPLE



VISUAL MEMORY ANCHOR

Constant acceleration turns displacement into rectangle + triangle.

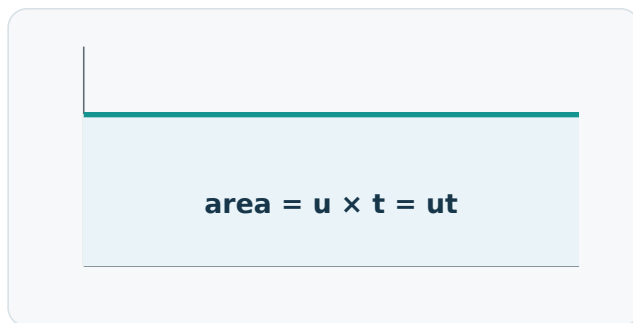
M-ST04B

Where the two terms come from

Rectangle gives ut ; triangle gives $\frac{1}{2}at^2$

WHY ut APPEARS

Imagine acceleration temporarily removed. If the object kept its initial velocity u for time t , the v - t graph would be a rectangle of height u and width t .



PHYSICAL MEANING

ut is the displacement the object would have covered from its starting velocity alone. The real v - t line rises above this rectangle; the extra region is a triangle.

WHY $\frac{1}{2}$ AND t^2 APPEAR

The triangle above the u -line has base t and height $v - u$.

$$\text{triangle area} = \frac{1}{2} \times t \times (v - u)$$

From $v = u + at$, the velocity increase is $v - u = at$.

$$\text{extra displacement} = \frac{1}{2} \times t \times at = \frac{1}{2}at^2$$

WHERE THE FACTORS COME FROM

$\frac{1}{2}$ comes from triangle area. One factor of t is the graph width; the other appears because the velocity increase grows as at . Thus $t \times at$ gives at^2 .

M-ST04B

Build and use the displacement relation

Put the two areas together, then interpret every term

$$s = ut + \frac{1}{2}at^2$$

rectangle + triangle

MEANING OF THE TERMS

s: total displacement during the interval.
 ut: part produced by the initial velocity.
 $\frac{1}{2}at^2$: extra part produced because acceleration changes the velocity during the interval.

WORKED EXAMPLE

A scooter is moving at 4 m/s and accelerates uniformly at 2 m/s² for 3 s. Find its displacement. Predict first: it must exceed 4 × 3 = 12 m because acceleration adds extra positive area.

SOLVE AND INTERPRET

$$s = ut + \frac{1}{2}at^2$$

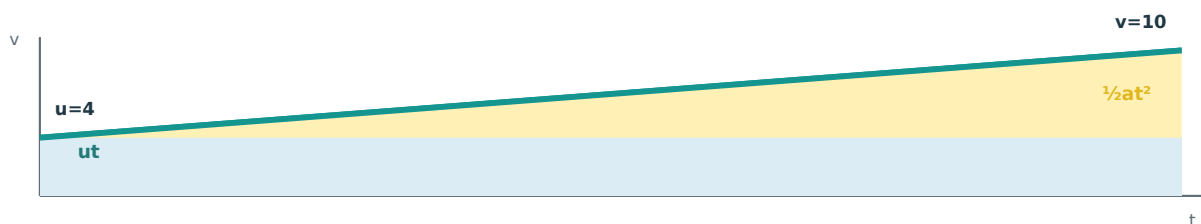
$$s = (4)(3) + \frac{1}{2}(2)(3^2)$$

$$s = 12 + 9 = 21 \text{ m}$$

AREA STORY + CHECK

12 m is the rectangle from the starting velocity. 9 m is the triangular extra displacement created by acceleration. 21 m is greater than 12 m, matching the prediction that the object speeds up during the interval.

SEE THE SAME RESULT AS AREA



M-ST04B

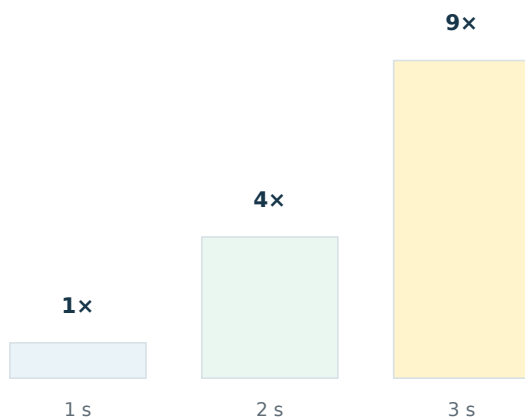
Special case and misconception

Starts from rest; final velocity $\times$ time is not generally displacement

STARTS FROM REST

Set $u = 0$. The rectangle disappears. Only the acceleration-created triangle remains.

$$s = \frac{1}{2}at^2$$



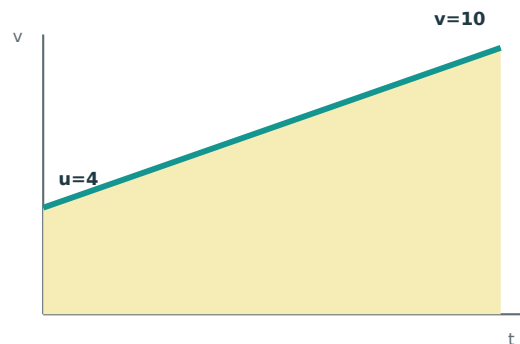
SCALING HELPER

For the same acceleration: double the time $\rightarrow$ four times the displacement; triple the time $\rightarrow$ nine times the displacement. Cumulative displacement after 1, 2, 3 s scales as 1:4:9. Do not confuse this with successive one-second distances 1:3:5.

MISCONCEPTION CLINIC

WRONG IDEA

"The final velocity is 10 m/s for a 3 s journey, so displacement is $10 \times 3 = 30$ m." This treats the object as though it moved at 10 m/s for the entire interval.



WHY IT FAILS + RETRY

When velocity changes, displacement is not simply final velocity $\times$ time. For $u = 4$ m/s, $a = 2$ m/s², $t = 3$ s, the true displacement is 21 m, not 30 m. If $u = 0$ and a is constant, compare vt with $\frac{1}{2}at^2$: final- $v \times t$ is exactly twice the true displacement.

M-ST04B

Application and model gate

Same equation with negative a - but only for one constant- a interval

A BUS COMES TO REST

A bus travels at 20 m/s and comes to rest uniformly in 4 s. Find a first: $0 = 20 + 4a$, so $a = -5 \text{ m/s}^2$.

$$s = (20)(4) + \frac{1}{2}(-5)(4^2)$$

$$s = 80 - 40 = 40 \text{ m}$$

PHYSICAL CHECK

The bus slows from 20 to 0 m/s, so its average speed must be less than 20 m/s. 40 m in 4 s corresponds to 10 m/s average speed - sensible. The negative triangle subtracts from the starting-velocity rectangle.

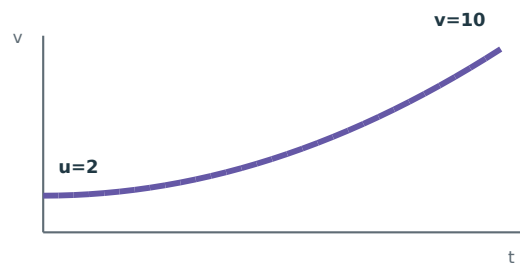
WHEN NOT TO USE $s = ut + \frac{1}{2}at^2$ AS ONE STEP

VALID

One interval with constant acceleration and a consistent sign convention.

NOT VALID AS ONE STEP

Acceleration changes during the interval, or the journey contains distinct stages such as reaction + braking.



GRAPH TEST

Straight v-t segment → constant a is plausible.
Curved v-t → acceleration changes; use a different or piecewise model.

M-ST04B

Guided practice

Full support on the left; faded support on the right

Guided 1 - full scaffold

A trolley starts at 3 m/s and accelerates uniformly at 2 m/s² for 5 s. Find its displacement.

$$s = ut + \frac{1}{2}at^2$$

Known: $u = 3$, $a = +2$, $t = 5$, $s = ?$. Final velocity is not needed.

PREDICT FIRST

Should the displacement be greater or less than $3 \times 5 = 15$ m?

H1: final velocity is not needed.
H2: one constant- a interval with u , a , t and target s .
H3: write $s = ut + \frac{1}{2}at^2$; do the substitution yourself.

Guided 2 - faded

A cart starts from rest and accelerates uniformly at 3 m/s² for 4 s. Find its displacement.

ONE CUE ONLY

Starts from rest $\rightarrow u = 0$. The rectangle disappears; the triangle remains.

U, A, T, S

WHICH AREA DISAPPEARS?

FIRST RELATION + CALCULATION

M-ST04B

Independent transfer

Choose your own representation, model gate, and relation

NOW YOU TRY - NO HELPER

A car moves at 8 m/s and accelerates uniformly at 1.5 m/s^2 for 6 s. Without drawing a supplied template, find the displacement and explain which part came from the starting velocity.

DRAW / CHOOSE A REPRESENTATION

STATE THE CONSTANT-A MODEL GATE

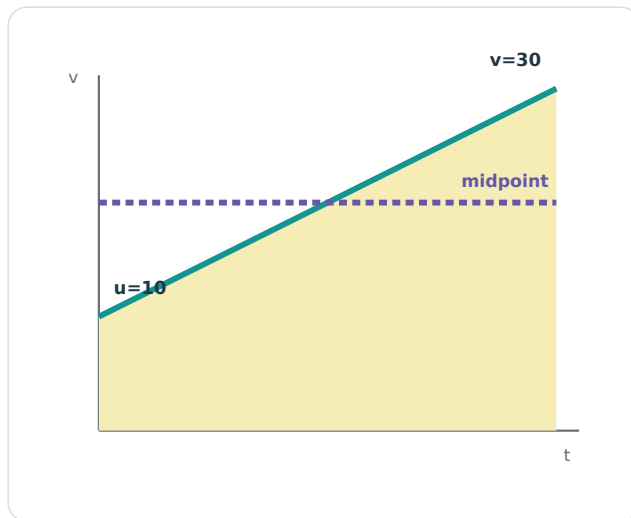
SOLVE FOR DISPLACEMENT

EXPLAIN THE TWO TERMS PHYSICALLY

M-ST04C

Average velocity under constant acceleration

A linear velocity ramp has a midpoint average - but only after the model gate



A CAR SPEEDS UP STEADILY

A car's velocity rises steadily from 10 m/s to 30 m/s over 5 s. Because acceleration is constant, velocity passes through the intermediate values at a uniform rate.

WHAT DO YOU EXPECT?

The car spends equal amounts of time below and above 20 m/s in a symmetric linear ramp. That makes 20 m/s a natural candidate for the average velocity.

MEMORY ANCHOR

LINEAR RAMP → MIDPOINT AVERAGE. This is a constant-acceleration result, not the universal definition of average velocity.

ALWAYS TRUE

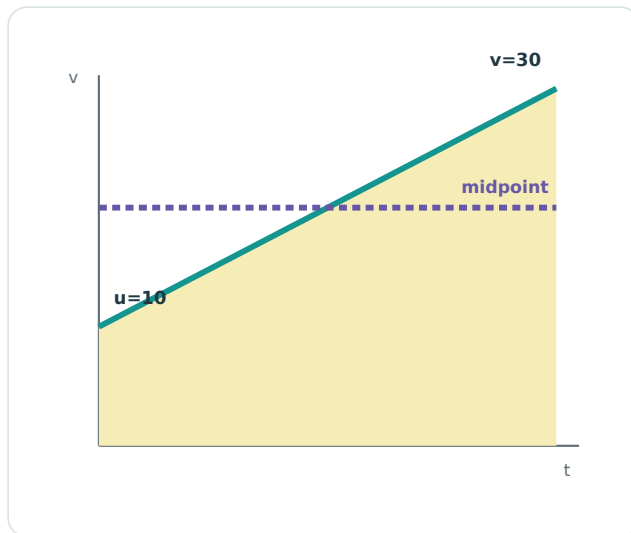
average velocity = displacement / elapsed time.
This remains the definition whether acceleration is constant, variable, or zero.

average velocity = displacement / time

M-ST04C

Why the midpoint is the average

See the straight ramp, then compress it



VISUAL IDEA

For a straight ramp, the part below the midpoint and the part above the midpoint balance. The mean height of the ramp is halfway between the endpoint heights.

$$\bar{v} = (u + v) / 2$$

constant acceleration → linear v-t ramp

FROM AVERAGE VELOCITY TO DISPLACEMENT

Start with displacement = average velocity × time. For a linear velocity ramp, replace average velocity by the midpoint of u and v .

$$s = ((u + v) / 2)t$$

WHAT DISAPPEARED?

Acceleration does not appear explicitly because its effect is already encoded in how far apart u and v are over the interval.

USEFUL VARIABLE PATTERN

If u , v and t are known and acceleration is not needed, this is usually the cleanest constant-acceleration relation. Check constant acceleration first; do not use endpoint average merely because u and v are given.

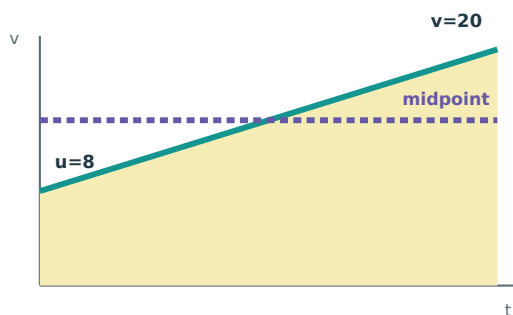
M-ST04C

Worked example and reusable helper

Reason first, compress second

QUESTION

A cyclist's velocity increases uniformly from 8 m/s to 20 m/s in 6 s. Find the displacement during this interval.



PREDICT

The average velocity must lie between 8 and 20 m/s. Because the ramp is linear, it should be exactly the midpoint.

$$\bar{v} = (8 + 20) / 2 = 14 \text{ m/s}$$

$$s = 14 \times 6 = 84 \text{ m}$$

CONCEPT HELPER: LINEAR RAMP → MIDPOINT

USE THIS HELPER WHEN YOU SEE...

Use this when: uniform acceleration; u and v are known; time is known; displacement is asked; one constant- a interval from start to finish.

THREE-LINE ROUTE

1. MODEL: constant $a \rightarrow v$ changes linearly.
2. AVERAGE: $\bar{v} = (u + v)/2$.
3. DISPLACEMENT: $s = \bar{v} \times t$.

WHY THIS HELPS

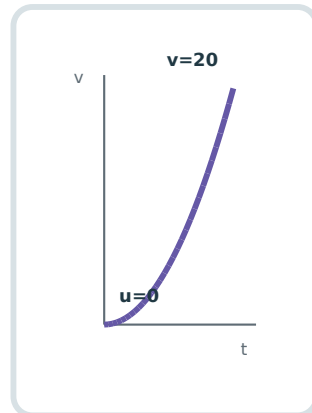
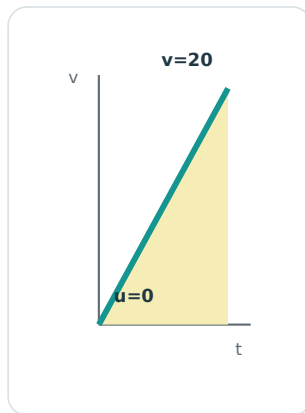
It keeps the model, the average, and the displacement step separate instead of turning the final relation into another isolated formula.

M-ST04C

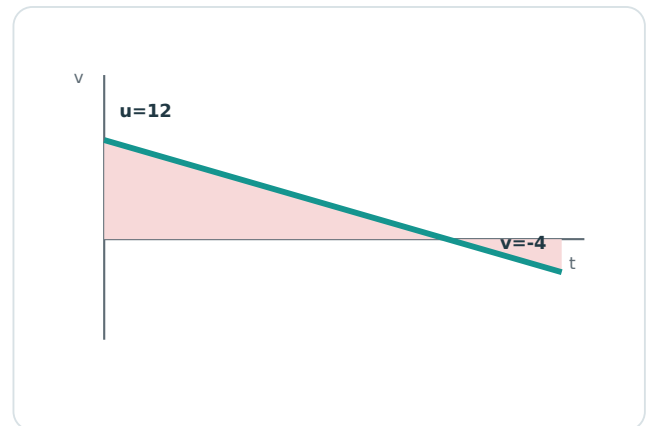
Two important limits

Endpoint average is not universal; displacement is not always distance

SAME ENDPOINTS CAN HIDE DIFFERENT HISTORIES



SIGNED VELOCITY STILL MATTERS



CORRECT MODEL

$(u + v)/2$ works for a linear v-t graph. A curved graph can spend more of the interval at lower velocity, giving a smaller area even with the same endpoints.

CONSTANT ACCELERATION CAN CROSS $v = 0$

If velocity changes uniformly from +12 m/s to -4 m/s in 4 s, $\bar{v} = (12 + (-4))/2 = 4$ m/s and $s = 4 \times 4 = 16$ m. This is signed net displacement; total distance is larger.

ANTI-TRIGGER

If a question asks for distance and velocity changes sign, split the v-t area at the zero crossing and add absolute areas. Do not use signed displacement as distance.

M-ST04C

Multistage journeys and model gates

Use midpoint average only inside the stage where the ramp is linear

TRAIN WITH TWO STAGES

A train starts from rest, accelerates uniformly to 80 km/h during time t , then travels at 80 km/h for another $3t$.

STAGE 1 HELPER

Uniform acceleration from 0 to 80 means average speed during stage 1 is $(0 + 80)/2 = 40$ km/h.

COMBINE THE STAGE DISTANCES

$$s_1 = 40t$$

$$s_2 = 80(3t) = 240t$$

$$\text{average speed, total} = 280t / 4t = 70 \text{ km/h}$$

WHY THIS IS A VARIANT

The midpoint shortcut applies only to the first stage. The whole journey is multistage, so total average speed still requires total distance / total time.

THREE MODEL GATES

1. ACCELERATION IS NOT CONSTANT

A curved v-t graph means endpoint midpoint is not generally the time-average.

2. THE TRIP HAS MULTIPLE STAGES

Split first. Use the shortcut separately only on stages that each have constant acceleration.

3. DISTANCE IS ASKED AFTER REVERSAL

Average velocity gives displacement. If velocity changes sign, total distance requires absolute path/area accounting.

FAST MODEL CHECK

Ask: one interval? constant a ? displacement or distance? Only then choose $s = ((u + v)/2)t$.

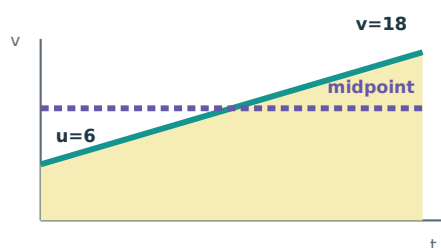
M-ST04C

Guided practice

Full support → faded support

Guided 1 - full scaffold

A car's velocity increases uniformly from 6 m/s to 18 m/s in 5 s. Find its displacement.



$$\bar{v} = (6 + 18) / 2 = 12 \text{ m/s}$$

$$s = \bar{v}t = 12 \times 5$$

Guided 2 - faded

A cart slows uniformly from 20 m/s to rest in 4 s. Find the displacement during braking.

ONE CUE

Rest means $v = 0$. Decide whether the midpoint-average shortcut is valid.

SKETCH THE V-T RAMP

MODEL

v

s

COMPARE THE TWO RAMPS

Both questions use the midpoint shortcut because the v-t graph is linear. Guided 1 ramps upward; Guided 2 ramps downward. The model gate is the same even though the acceleration sign differs.

M-ST04C

Independent transfer

Represent the sign change before choosing the quantity

NOW YOU TRY - NO SUPPLIED DIAGRAM

A trolley has velocity -6 m/s and constant acceleration. Four seconds later its velocity is $+10 \text{ m/s}$. Find the displacement. Then explain why your answer is not automatically the total distance travelled.

FIRST MOVE

Decide the sign convention, sketch a v-t ramp, identify the zero crossing qualitatively, and choose the correct quantity: displacement or distance.

SKETCH THE SIGNED V-T RAMP

FIND THE AVERAGE VELOCITY AND DISPLACEMENT

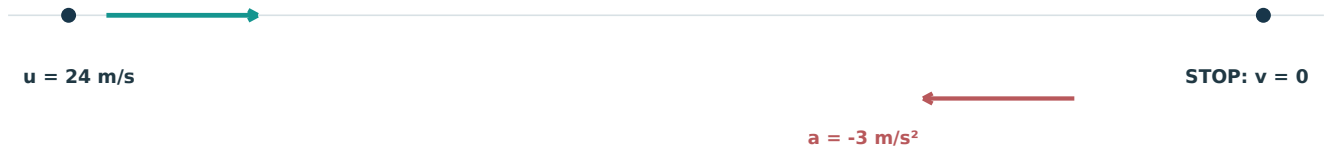
EXPLAIN WHY TOTAL DISTANCE MAY DIFFER

M-ST04D

Remove time: why velocities become squared

Use a time-free constant-acceleration relation when distance and velocity are the real targets

stopping distance $s = ?$



MEMORY ANCHOR

TIME ABSENT → THINK VELOCITY-DISTANCE. For one constant-acceleration interval, $v^2 = u^2 + 2as$ connects velocity directly to displacement.

REAL-LIFE REASON

A car is moving at 20 m/s and brakes with constant acceleration -4 m/s^2 . The stopping distance is asked; stopping time is not.

WHAT IS KNOWN?

VARIABLES

$u = 20 \text{ m/s}$, $v = 0$, $a = -4 \text{ m/s}^2$, $s = ?$. Time is the unnecessary variable.

WHY NOT START WITH $v = u + at$?

That relation introduces time. It would work in two steps, but a time-free relation connects the variables we actually care about in one step.

BIG QUESTION

Can we remove time from the constant-acceleration model without inventing a new law?

M-ST04D

Build the time-free relation

Eliminate time step by step - every transformation stays visible

DERIVATION LADDER

1 SOLVE THE VELOCITY RELATION FOR TIME

$$t = (v - u) / a$$

from $v = u + at$

2 START FROM DISPLACEMENT = AVERAGE VELOCITY × TIME

$$s = ((u + v) / 2)t$$

constant acceleration gives the midpoint average

3 SUBSTITUTE THE TIME EXPRESSION

$$s = ((u + v)/2)((v - u)/a)$$

elimination, not a new assumption

4 USE SUM × DIFFERENCE

$$2as = v^2 - u^2$$

because $(u + v)(v - u) = v^2 - u^2$

5 REARRANGE

$$v^2 = u^2 + 2as$$

time has disappeared

WHY THE SQUARES APPEAR

The factor $(u + v)$ comes from average velocity and $(v - u)$ comes from velocity change. Their product is a difference of squares.

WHY THIS MATTERS

You can rebuild the relation from earlier ideas. The squared velocities are a consequence of elimination, not a mysterious pattern to memorise.

M-ST04D

What changes evenly with distance?

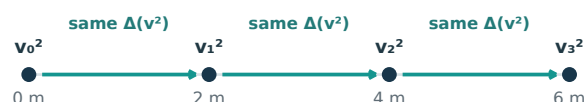
For constant acceleration, equal distance steps give equal changes in v^2

THE PHYSICAL MEANING

VELOCITY-DISTANCE RELATION

For constant acceleration, every displacement s changes v^2 by an amount $2as$. Velocity squared is the quantity that changes linearly with displacement.

EQUAL-DISTANCE HELPER



$$\Delta(v^2) = 2a \Delta s$$

GRADE 9 CHECK

Can you explain in words why equal displacement intervals under the same acceleration give equal changes in v^2 ?

RECOGNITION CUE

If a question gives equal distances under the same constant acceleration or retardation, compare changes in v^2 before reaching for time.

WHY THIS HELPS

Equal-distance questions can look awkward in time because each segment takes a different duration. Under constant retardation, v^2 loses the same amount over equal distances.

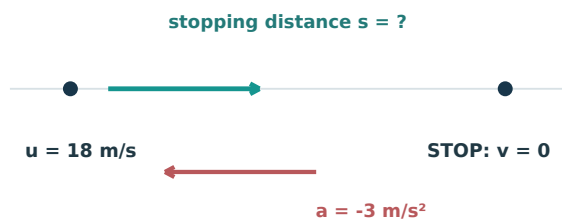
M-ST04D

Stopping distance

Use the time-free relation, then see the speed-squared scaling

WORKED EXAMPLE

A bicycle moves at 18 m/s and brakes uniformly with acceleration -3 m/s^2 . Find the stopping distance. Predict first: the distance must be positive; the negative sign belongs to acceleration.



$$0^2 = 18^2 + 2(-3)s$$

$$0 = 324 - 6s$$

$$s = 54 \text{ m}$$

PHYSICAL CHECK

A moderate deceleration needs tens of metres to remove an 18 m/s speed. 54 m is physically plausible.

STOPPING DISTANCE SCALES WITH SPEED SQUARED

$$0 = u^2 + 2as$$

$$\text{stopping distance} = u^2 / (2|a|)$$

SAME BRAKING CONDITION

If $|a|$ is unchanged, stopping distance is proportional to u^2 . Double $u \rightarrow$ four times the distance. Triple $u \rightarrow$ nine times the distance.

REAL-LIFE CONTRAST

Car A: 40 km/h $\rightarrow$ 40 m. Car B: 80 km/h under the same braking condition $\rightarrow 4 \times 40 = 160 \text{ m}$.

SAFETY INTERPRETATION

Higher speed is costly because stopping distance grows quadratically under the same braking acceleration.

M-ST04D

Repair a proportionality mistake

Then reuse one braking condition in a new situation

MISCONCEPTION CLINIC

WRONG IDEA

"If speed doubles, stopping distance should also double." That would require direct proportionality to speed, but stopping distance under fixed braking acceleration depends on speed squared.

CORRECT MODEL

Stopping distance $\propto u^2$. If $u \rightarrow 2u$, then $u^2 \rightarrow 4u^2$ and $s \rightarrow 4s$. Compare the square of the speed ratio when braking conditions are the same.

RETRY

If speed becomes 1.5 times as large, by what factor does the stopping distance change?

APPLICATION: A TRAIN BRAKES TOO LATE

THE STORY

A train at 20 m/s normally needs 500 m to stop with a fixed braking acceleration. If braking begins only 250 m before the station, it will still be moving at the station.

FIRST MOVE

Use the full 500 m stopping case to determine the shared braking acceleration. Then use the same v^2 relation again over 250 m.

HIDDEN STRUCTURE

The acceleration is not given directly, but the first stopping case determines it. The second case reuses the same a with a shorter displacement. This is the same relation used twice, not a new formula.

M-ST04D

Model gate + Guided 1

Choose the time-free route only when it matches the interval and target

WHEN NOT TO USE ONE TIME-FREE RELATION

1. ACCELERATION CHANGES

If resistance or acceleration changes substantially, one constant- a relation is not valid over the whole path.

2. SEPARATE STAGES

Reaction time + braking, different road surfaces, or different acceleration stages must be split first.

3. TIME IS ACTUALLY THE TARGET

The time-free relation may be true but inefficient. Use the relation that connects the target while avoiding unnecessary unknowns.

FAST CHECK

Ask: constant a ? one interval? distance/velocity target? time unnecessary? If yes, $v^2 = u^2 + 2as$ is a strong candidate.

GUIDED 1 - FULL SCAFFOLD

QUESTION

A scooter moving at 16 m/s brakes uniformly at 2 m/s². Find the stopping distance.

$$v^2 = u^2 + 2as$$

KNOWN

$u = 16$ m/s, $v = 0$, $a = -2$ m/s², $s = ?$. Time is not asked for.

HINTS - REVEAL PROGRESSIVELY

H1: stops $\rightarrow v = 0$; time is not asked for.

H2: one constant-acceleration braking interval.

H3: write $0 = 16^2 + 2(-2)s$.

M-ST04D

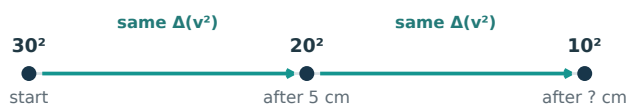
Guided 2 - equal-distance v^2 thinking

Faded support: infer the distance from repeated v^2 loss

QUESTION

A pellet enters a material at 30 m/s and experiences constant retardation. After travelling 5 cm its speed is 20 m/s. Without finding time, determine the extra distance needed to reduce its speed from 20 m/s to 10 m/s.

REPRESENTATION



ONE MODEL CUE

Constant retardation means the same $2a$ factor applies throughout. Compare changes in v^2 , not changes in v .

YOUR JOB

LOSS IN v^2 OVER FIRST 5 CM

LOSS IN v^2 FOR 20 → 10

DISTANCE RATIO / EXTRA DISTANCE

HINTS - REVEAL ONLY IF STUCK

H1: same constant retardation → same $2a$.

H2: compare changes in v^2 .

H3: first loss = $30^2 - 20^2 = 500$ over 5 cm.

M-ST04D

Independent transfer

Time never needs to appear

NOW YOU TRY - NO SUPPLIED DIAGRAM

A train of length L accelerates uniformly. Its engine passes a signal at speed u and its last carriage passes the same signal at speed v . Find the speed when the midpoint of the train passes the signal.

FIRST MOVE

Recognise that the midpoint corresponds to half the train-length displacement. Use the velocity-distance relation twice; time never needs to appear.

DRAW THE TRAIN / SIGNAL AND MARK THE TWO KNOWN SPEEDS

WRITE THE v^2 RELATION FOR THE FULL LENGTH L

WRITE THE SAME RELATION FOR HALF-LENGTH $L/2$

ELIMINATE THE SHARED ACCELERATION AND FIND THE MIDPOINT SPEED

M-ST04E

Read a motion equation from its coefficients

Match the quadratic position-time expression to the standard constant-acceleration form

$$x(t) = x_0 + ut + \frac{1}{2}at^2$$

CONSTANT TERM

 x_0

initial position

COEFFICIENT OF T

 u

initial velocity

COEFFICIENT OF T^2 **$\frac{1}{2}a$** so $a = 2 \times \text{coefficient}$

MEMORY ANCHOR

Match coefficients to $x(t) = x_0 + ut + \frac{1}{2}at^2$ before doing algebra. The constant gives x_0 , the coefficient of t gives u , and the coefficient of t^2 gives $a/2$.

EXAMPLE

$x = 5t^2 - 4t + 5 \rightarrow x_0 = 5 \text{ m}, u = -4 \text{ m/s}, a = 10 \text{ m/s}^2$. The negative u tells the initial direction.

THEN USE THE READ-OFF VALUES

If velocity is asked, use $v = u + at$ after identifying u and a . Read the equation first; only then choose the next relation.

M-ST04E

Boundary condition + Guided 1

The coefficient shortcut works only for the constant-acceleration quadratic form

MISCONCEPTION CLINIC

WRONG IDEA

"Any $x(t)$ polynomial can be read using $x = x_0 + ut + \frac{1}{2}at^2$."

REPAIR

The shortcut applies when the position equation is exactly quadratic in time and represents constant acceleration. A t^3 term signals a different model.

CHECK

After reading a , verify that it is constant and that the units match.

GUIDED 1 - FULL SUPPORT

QUESTION

$x(t) = 3 + 6t + 2t^2$. Find initial position, initial velocity and acceleration.

$$x(t) = x_0 + ut + \frac{1}{2}at^2$$

READ → TRANSLATE

constant term $\rightarrow x_0$
 coefficient of $t \rightarrow u$
 coefficient of $t^2 \rightarrow a/2$

HINTS

H1: compare the standard shape.
 H2: $u = 6$.
 H3: $\frac{1}{2}a = 2 \rightarrow a = 4$.

M-ST04E

Guided 2 + transfer

Read the coefficients first; then choose what the question needs

GUIDED 2 - FADED

QUESTION

$x(t) = 7 - 5t + 4t^2$. Find the velocity at $t = 3$ s without differentiating.

ONE CUE

Read u and a from the quadratic form, then use $v = u + at$.

U

A

V(3)

HINT ONLY IF STUCK

$u = -5$ and $a = 8$; then use $v = u + at$.

TRANSFER

TASK

Given a quadratic position-time expression, first identify x_0 , u and a . Only then decide whether the question asks position, velocity, acceleration or a meeting condition.

ANTI-TRIGGER

If a t^3 or other non-quadratic term appears, do not silently force the constant-acceleration coefficient map.

NO CALCULUS NEEDED HERE

For this source-level constant-acceleration pattern, coefficient reading plus $v = u + at$ is enough when velocity is asked.

CREATE YOUR OWN QUADRATIC EXAMPLE AND LABEL x_0 , u AND a

M-ST05A

Distance during the n th second

The n th second is an interval, not the total time from the start



MEMORY ANCHOR

“During the n th second” means the interval $(n-1) \rightarrow n$. Start with cumulative displacement and subtract the two endpoints.

FIRST RECOGNITION CUE

“In 5 seconds” means the whole interval $0 \rightarrow 5$ s.
“During the 5th second” means only $4 \rightarrow 5$ s.
Interval language includes “during the 7th second”, “in the n th second”, and “next t seconds”.

TWO DIFFERENT QUESTIONS

QUESTION A

How far does the object travel in 5 seconds? $\rightarrow$ use the whole interval $0 \rightarrow 5$ s.

QUESTION B

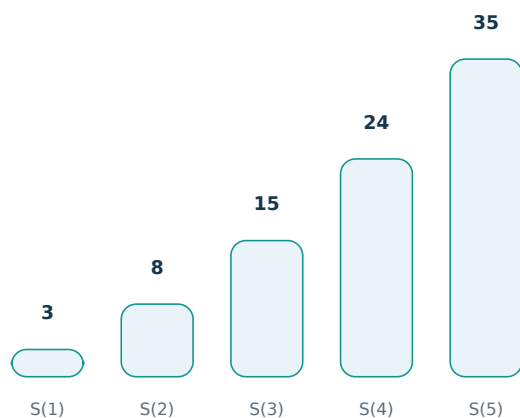
How far does the object travel during the 5th second? $\rightarrow$ isolate the one-second slice $4 \rightarrow 5$ s.

M-ST05A

Cumulative displacement is a running total

Later total - earlier total gives the interval displacement

THINK LIKE A TRIP METER



REMOVE WHAT CAME BEFORE

$S(5)$ contains everything from 0 to 5 s. $S(4)$ contains everything from 0 to 4 s. The displacement during the fifth second is the new total minus the old total.

UNIVERSAL INTERVAL HELPER

interval displacement = $S(\text{end}) - S(\text{start})$

$$s_5 = S(5) - S(4)$$

WHY THIS IS BROADER THAN A SHORTCUT

This subtraction idea works for any interval. The nth-second formula is only a constant-acceleration one-second shortcut built from this general idea.

M-ST05A

Derive the nth-second relation

Start from $S(t)$, substitute the two interval endpoints, then simplify

DERIVATION LADDER

$$S(t) = ut + \frac{1}{2}at^2$$

cumulative displacement for constant acceleration

$$s_n = S(n) - S(n-1)$$

the nth second runs from $n-1$ to n

$$s_n = [un + \frac{1}{2}an^2] - [u(n-1) + \frac{1}{2}a(n-1)^2]$$

substitute both cumulative totals

$$s_n = u + (a/2)(2n - 1)$$

simplified one-second interval relation

WHAT THE FORMULA MEANS

Every later one-second interval begins with a different velocity. Under constant acceleration, the interval displacement changes in a regular arithmetic pattern.

WHEN TO STEP BACK

The shortcut is derived from cumulative subtraction. If the interval is not one second, acceleration is not constant, or distance is asked across a reversal, return to $S(\text{end}) - S(\text{start})$ and the correct motion model.

M-ST05A

Worked example: the 4th second

Use the timeline first, then choose either subtraction or the shortcut

QUESTION

A particle has $u = 2 \text{ m/s}$ and constant acceleration $a = 3 \text{ m/s}^2$. Find its displacement during the 4th second.

4th second = 3 → 4 s



FIRST MOVE

Do not put $t = 4$ into $S(t)$ and stop. That gives the total from 0 to 4 s, not the displacement during the 4th second.

TWO EQUIVALENT ROUTES

ROUTE A - SUBTRACT

Compute $S(4) - S(3)$. $S(4) = 32 \text{ m}$ and $S(3) = 19.5 \text{ m}$, so $s_4 = 12.5 \text{ m}$. This is the most transferable method.

ROUTE B - SHORTCUT

$$s_4 = u + (a/2)(2 \times 4 - 1) = 2 + 1.5 \times 7 = 12.5 \text{ m}.$$

REUSABLE THREE-STEP ROUTE

1 MARK THE CLOCK: write the actual interval. 2 WRITE THE TOTAL: use $S(t)$. 3 SUBTRACT: later total - earlier total. Memory phrase: TOTAL → TOTAL → DIFFERENCE.

M-ST05A

Repair the interval mistake

Then use two separated nth-second distances to reveal acceleration

MISCONCEPTION CLINIC

WRONG IDEA

"The question says 5th second, so I put $t = 5$ into $S(t)$." That calculates the total accumulated from the start up to 5 s.

what the 5th second contains



CORRECT MODEL

The 5th second is one one-second slice: $4 \rightarrow 5$ s. Therefore use $S(5) - S(4)$. Retry: what interval does "during the 11th second" mean?

TWO NTH-SECOND DISTANCES REVEAL A

$$s_n = u + (a/2)(2n - 1)$$

$$s_{n+2} = u + (a/2)(2n + 3)$$

$$s_{n+2} - s_n = 2a$$

u and n cancel

NUMERICAL ILLUSTRATION

$115.0 - 102.5 = 12.5 = 2a$, so $a = 6.25 \text{ m/s}^2$. The arithmetic is easy; the challenge is recognising two interval displacements and choosing subtraction that cancels hidden quantities.

M-ST05A

A close variant: the next t seconds

The interval idea is the same even when the interval width changes

QUESTION

A toy starts from rest and travels 10 m in the first t seconds. How far does it travel in the next t seconds?



DO NOT READ “NEXT” AS TOTAL

The second piece is $S(2t) - S(t)$, not $S(2t)$.

USE THE CUMULATIVE LAW

$S(t) = \frac{1}{2}at^2 = 10$. $S(2t) = \frac{1}{2}a(2t)^2 = 40$.
Therefore the next t seconds contribute $40 - 10 = 30$ m.

WHEN THE NTH-SECOND SHORTCUT IS NOT SAFE

1. ACCELERATION IS NOT CONSTANT

Use $S(\text{end}) - S(\text{start})$ with the correct model; the compact nth-second relation assumes constant acceleration.

2. INTERVAL IS NOT ONE SECOND

For $2.5 \rightarrow 4.0$ s use $S(4.0) - S(2.5)$. Do not force the nth-second shortcut.

3. DISTANCE AFTER REVERSAL

$S(\text{end}) - S(\text{start})$ gives signed displacement. If distance is asked and the object reverses, split at the turning point.

FAST CHECK

Constant a? One-second interval? No reversal if distance is asked? If yes, the compact nth-second formula is safe.

M-ST05A

Guided practice

Full support on the left, faded support on the right

GUIDED 1 - FULL SCAFFOLD

A particle starts with $u = 4 \text{ m/s}$ and constant acceleration 2 m/s^2 . Find the displacement during the 6th second.

6th second = $5 \rightarrow 6 \text{ s}$ 

$$s_6 = s(6) - s(5)$$

operative relation is visible

HINTS

H1: "6th second" means $5 \rightarrow 6 \text{ s}$. H2: constant acceleration means $S(t) = ut + \frac{1}{2}at^2$ is valid. H3: start from $s_6 = S(6) - S(5)$.

GUIDED 2 - FADED SUPPORT

A uniformly accelerated body travels 54 m during its n th second and 66 m during its $(n+2)$ th second. Find the acceleration.

WRITE S_n

WRITE S_{n+2}

SUBTRACT AND SOLVE

HINTS - REVEAL PROGRESSIVELY

H1: the intervals are two seconds apart. H2: $S_{n+2} - S_n = 2a$. H3: $66 - 54 = 2a$.

M-ST05A

Independent transfer

Build the intervals yourself before calculating

TASK

A body starts from rest with constant acceleration. It covers 18 m in the first 3 s. Without using the nth-second shortcut directly, find the displacement during the next 3 s.

YOUR FIRST MOVE

No timeline is supplied. Build the two intervals $0 \rightarrow 3$ s and $3 \rightarrow 6$ s. Use cumulative displacement and subtract.

DRAW THE TWO TIME INTERVALS

WRITE THE CUMULATIVE DISPLACEMENT MODEL

FIND $s(3)$, THEN $s(6)$

SUBTRACT THE NEXT 3-SECOND INTERVAL AND EXPLAIN

M-ST05B

Why successive distances follow 1 : 3 : 5 : 7

From rest + constant acceleration + equal successive time intervals

1 parts

1st

3 parts

2nd

5 parts

3rd

7 parts

4th

MEMORY ANCHOR

SQUARE TOTALS → ODD DIFFERENCES.
Cumulative factors 1, 4, 9, 16 produce
successive interval factors 1, 3, 5, 7.

THREE CONDITIONS

The pure odd-number ratio requires: starts from rest, constant acceleration, and successive equal-duration time blocks.

A TOY CAR FROM REST

PHYSICAL STORY

Let $a = 2 \text{ m/s}^2$, so $S(t) = t^2$ metres. The cumulative totals are 1, 4, 9, 16 m after 1, 2, 3, 4 s. The successive one-second additions are 1, 3, 5, 7 m.

WHY THE GAPS GET BIGGER

The car enters each new second with a larger velocity. Equal-duration intervals therefore contain larger average velocities. With $a = 2 \text{ m/s}^2$, the interval average velocities are 1, 3, 5, 7 m/s.

M-ST05B

Square totals become odd differences

Two derivations show the same motion from different angles

ROUTE 1 - FROM CUMULATIVE TOTALS

$$S(t) = \frac{1}{2}at^2$$

from rest

IGNORE THE COMMON FACTOR

For comparison, every cumulative displacement contains the same factor $a/2$. The time-squared part creates the pattern.

FIRST DIFFERENCES

$1^2, 2^2, 3^2, 4^2 \rightarrow 1, 4, 9, 16$. Then $1-0=1, 4-1=3, 9-4=5, 16-9=7$.

ROUTE 2 - FROM THE NTH-SECOND RELATION

$$s_n = u + (a/2)(2n - 1)$$

$$u = 0 \rightarrow s_n = (a/2)(2n - 1)$$

$$s_n \propto 2n - 1$$

$$n = 1, 2, 3, 4 \rightarrow 1, 3, 5, 7$$

WHY TWO DERIVATIONS HELP

One route comes from square cumulative totals; the other comes from the nth-second formula. They must agree because both describe the same motion.

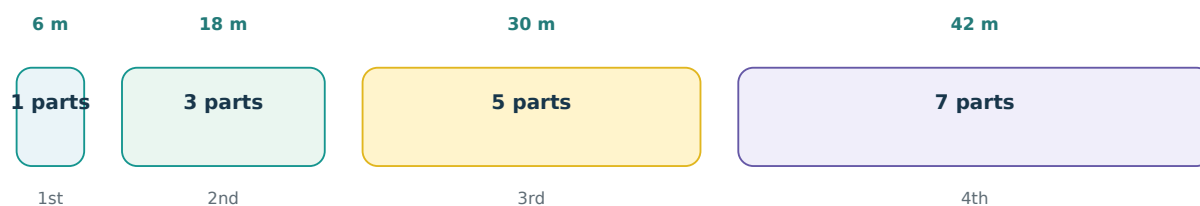
M-ST05B

Use the odd-number pattern efficiently

Let the first interval set the scale, then cross-check with cumulative totals

PATTERN HELPER

If the first equal-time interval displacement is d , then successive intervals are $d, 3d, 5d, 7d, \dots$. The intervals may be 1 s, 2 s, or any equal width Δt , provided the motion starts from rest and acceleration is constant.



WORKED EXAMPLE

A body starts from rest with constant acceleration and covers 6 m during the first second. The fourth-second factor is 7, so $s_4 = 7 \times 6 = 42$ m. A later interval should indeed be much larger.

RECOVER THE TOTAL FROM ONE INTERVAL

If the fourth-second displacement is 14 m, then $7d=14$, so $d=2$ m. After four equal intervals the cumulative factor is $4^2=16$, hence total displacement = $16d = 32$ m.

M-ST05B

The 1 : 3 : 5 : 7 ratio is not universal

Three conditions must hold before using the shortcut

MISCONCEPTION CLINIC

WRONG IDEA

"Under constant acceleration, successive-second distances are always 1:3:5:7." Constant acceleration alone is not enough.

COUNTEREXAMPLE

Let $u = 10 \text{ m/s}$ and $a = 2 \text{ m/s}^2$. Successive one-second displacements are 11, 13, 15, 17 m - an arithmetic sequence, but not the pure odd-number ratio.

CORRECT MODEL

The pure ratio requires $u=0$, constant acceleration, and successive equal time intervals measured from the start. Retry: would it apply to a car already moving at 8 m/s ?

WHEN THE SHORTCUT FAILS

1. INITIAL VELOCITY IS NOT ZERO

Use the full s_n relation; nonzero u shifts every interval displacement.

2. ACCELERATION CHANGES

There is no fixed odd pattern because the cumulative displacement is no longer one quadratic law across the whole interval.

3. TIME BLOCKS ARE UNEQUAL

For unequal widths, evaluate $S(\text{end}) - S(\text{start})$ directly. The 1:3:5:7 pattern belongs to successive equal-duration blocks.

FAST GATE

Starts from rest? Constant a ? Equal successive time blocks? If all three are yes, the odd-number pattern is available.

M-ST05B

Guided practice + independent transfer

Use the pattern only after the model gate passes

GUIDED 1 - FULL SCAFFOLD

A particle starts from rest under constant acceleration. It covers 3 m in the first second. Find the displacement during the fourth second.



HINTS

H1: starts from rest + constant acceleration + equal 1 s blocks. H2: 1st : 4th = 1 : 7. H3: $s_4 = 7 \times 3$.

GUIDED 2 - FADED SUPPORT

A body starts from rest with constant acceleration. It travels 20 m during the third second. Find the displacement during the fifth second.

3RD-SECOND FACTOR

5TH-SECOND FACTOR

RATIO / ANSWER

INDEPENDENT TRANSFER

A trolley starts from rest with constant acceleration. During the first 2 s it covers 8 m. Find how far it travels during the next 2 s. Solve once using the equal-interval odd-number pattern and once using cumulative displacement.

YOUR FIRST MOVE

No diagram is supplied. The equal intervals are 2 s wide, not 1 s. Decide whether the same 1:3 pattern applies and justify it.

DRAW TWO EQUAL 2 S INTERVALS

PATTERN ROUTE

CUMULATIVE-DISPLACEMENT ROUTE

COMPARE AND JUSTIFY

M-ST06A

Delayed start: one global clock, different elapsed times

Same observation instant does not mean the same motion duration

A active: elapsed time = T



MEMORY ANCHOR

GLOBAL TIME $\rightarrow$ LOCAL ELAPSED TIME. If A has moved for T and B starts τ later, B has moved for $T - \tau$.

HIDDEN FACT

At global time $T=6$ s with a 2 s delay, A has moved for 6 s while B has moved for only 4 s. Translate the times before any kinematics.

TRANSLATE THE CLOCK BEFORE WRITING EQUATIONS

GLOBAL EVENT

Observe both bodies at the same global instant T .

BODY A

Starts at the global origin, so elapsed time = T .

BODY B

Starts τ later, so elapsed time = $T - \tau$.

M-ST06A

Why using the same t for both bodies fails

Repair the clock mismatch before writing motion equations

MISCONCEPTION CLINIC

WRONG IDEA

"They are compared at the same instant, so both use the same t ." The t in a motion equation means elapsed time since that body or stage began.

WHY IT FEELS PLAUSIBLE

Both positions are evaluated at one global instant, so it is easy to confuse clock time with motion duration.

CORRECT MODEL

Use one global clock to locate the event, then convert that instant into each body's own elapsed time.

GLOBAL → LOCAL → EQUATION

$$t_A = T$$

$$t_B = T - \tau$$

THREE-STEP HELPER

1 GLOBAL: choose one clock for the whole story.
2 LOCAL: convert the global instant into each body's elapsed time. 3 EQUATION: only now write each motion equation.

MEETING / MULTISTAGE WARNING

At a meeting, positions are equal at one global instant, but local times may differ. If a body changes stage, mark the stage boundary and reset local elapsed time for that stage.

M-ST06A

Same acceleration, different start times

Translate the times first; then look for cancellation

BOTH START FROM REST WITH THE SAME A

$$S_A = \frac{1}{2}aT^2$$

$$S_B = \frac{1}{2}a(T-\tau)^2$$

RATIO SHORTCUT

The common factor $a/2$ cancels, so displacement ratios reduce to squared elapsed-time ratios.

EXAMPLE

A starts now. B starts 2 s later. At global time 6 s, the displacement ratio A:B is $6^2:4^2 = 36:16 = 9:4$.

GUIDED

TASK

Car A starts from rest at $t=0$ with constant acceleration a . Car B starts from the same point 3 s later, also from rest with the same acceleration. At global time T , write the ratio of A's displacement to B's displacement.

HINTS

H1: mark A from 0 to T . H2: mark B from 3 to T . H3: use T and $T-3$ inside the same $\frac{1}{2}at^2$ model.

GLOBAL CLOCK

LOCAL ELAPSED TIMES

RATIO

M-ST06A

Independent transfer

Build the global clock yourself before solving

TASK

Two identical carts start from the same point under the same constant acceleration. The second starts 4 s after the first. At some global instant, the first has travelled four times as far as the second. Find the elapsed time of the first cart.

MODEL GATE

This ratio shortcut assumes both carts start from rest with the same constant acceleration. No diagram is supplied: let the first elapsed time be T and the second $T-4$.

DRAW ONE GLOBAL CLOCK AND MARK BOTH STARTS

WRITE EACH LOCAL ELAPSED TIME

WRITE THE TWO DISPLACEMENT MODELS

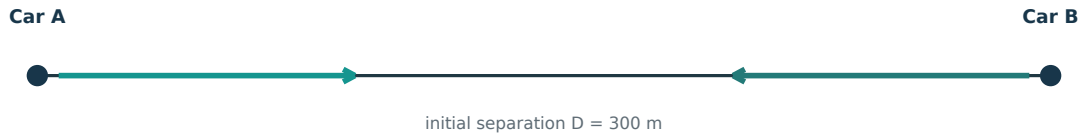
USE THE 4:1 DISTANCE CONDITION

SOLVE AND CHECK THE RESULT AGAINST THE 4 S DELAY

M-ST06B

Two braking cars: kinematics, then geometry

Stop each car independently; combine the path lengths afterward



STAGE 1 - EACH CAR STOPS

For each car, use its own initial speed and braking magnitude to find its stopping distance. These are two independent kinematics calculations.

STAGE 2 - COMPARE WITH THE GAP

Only after both stopping distances are known do you ask whether the cars collide, just touch, or stop with a positive gap.

$$\text{remaining gap} = D - (s_1 + s_2)$$

positive = gap • zero = just stop together • negative = collision before stop

GAP > 0

Both cars stop before using the full separation; a positive gap remains.

GAP = 0

The stopping paths exactly use the initial separation: just-touch threshold.

GAP < 0

Their required stopping paths exceed D , so collision occurs before both have stopped.

M-ST06B

Stopping-distance helper

One car at a time, then the geometry layer

$$s_{\text{stop}} = u^2 / (2|a|)$$

compact form for one uniformly braking vehicle

WHY |A|?

Stopping distance is a positive path length. The acceleration points opposite the motion, so use the deceleration magnitude in this compact path-length form.

TWO-CAR TEMPLATE

$$s_1 = u_1^2 / (2|a_1|)$$

$$s_2 = u_2^2 / (2|a_2|)$$

$$\text{gap} = D - s_1 - s_2$$

WORKED EXAMPLE: WILL THEY COLLIDE?

QUESTION

Two cars approach each other from 300 m apart. Each moves at 20 m/s and each brakes uniformly at 2 m/s². Find the gap when both stop.

SOLUTION

$$s_1 = s_2 = 20^2 / (2 \times 2) = 100 \text{ m. Then gap} = 300 - 100 - 100 = 100 \text{ m.}$$



PHYSICAL CHECK

The combined stopping path is 200 m, less than the 300 m separation, so a positive gap must remain.

M-ST06B

Do not mix signs and geometry

Keep coordinate sign logic separate from the final path-length comparison

WRONG IDEA

“One car moves right and one left, so one stopping distance should be negative.” That mixes signed displacement with the positive path length each car consumes from the separation.

REPAIR

Compute each stopping path length as a positive number, then subtract their sum from the initial separation. Signed coordinates can also work, but the convention must remain consistent from equations to interpretation.

WHY IT FEELS PLAUSIBLE

A single coordinate axis gives opposite signs to the cars’ displacements, but the final collision/gap test asks how much geometric separation has been used up.

GEOMETRY LAYER AND VARIANTS

MINIMUM SAFE SEPARATION

The smallest D that avoids collision is $s_1 + s_2$. Equality is the just-touch threshold.

DIFFERENT BRAKES

If one car is faster but also brakes harder, do not guess which travels farther. Compare $u^2/|a|$ for each car.

RECOGNITION CUE

Two moving objects + braking + known initial gap
→ solve stopping paths independently, then combine.

COORDINATE-METHOD NOTE

A signed-coordinate solution can also work, but keep the coordinate convention and final geometric interpretation consistent from start to finish.

M-ST06B

Guided practice + independent transfer

Solve the physics layer first, then the geometry layer

GUIDED

Cars A and B approach each other from 250 m apart. A: $u=18$ m/s, $|a|=3$ m/s². B: $u=12$ m/s, $|a|=2$ m/s². Find the final gap if both brake immediately.

INDEPENDENT TRANSFER

Two vehicles approach each other. Their stopping distances under current road conditions are 72 m and 98 m. What is the minimum initial separation that guarantees they stop without collision?

$$s = u^2 / (2|a|)$$

use once for A and once for B

HINTS

H1: solve A alone. H2: solve B alone. H3: subtract the two stopping distances from 250 m.

RECOGNITION CUE

No kinematics is needed. This is the geometry layer only: the minimum safe separation equals the sum of the two stopping paths.

STOPPING DISTANCE A

STOPPING DISTANCE B

REMAINING GAP + INTERPRETATION

COMBINE THE TWO PATH LENGTHS

STATE THE MINIMUM SAFE D

EXPLAIN WHY EQUALITY IS THE THRESHOLD

M-ST06C

Velocity change + average velocity: a short route

Definitions can be cleaner than reconstructing every SUVAT variable

TWO-STEP ROUTE

USE THIS WHEN YOU KNOW

Constant acceleration a , total velocity change Δv , and average velocity over the same interval.

MODEL GATE

Δv , a and average velocity must all refer to the same interval. If they come from different stages, split the motion first.

$$t = \Delta v / a$$

velocity change $\div$ rate of change = time

UNITS TELL THE STORY

$$(\text{m/s}) / (\text{m/s}^2) = \text{s}$$

$$s = v_{\text{avg}} t$$

average velocity $\times$ time = displacement

$$(\text{m/s}) \times \text{s} = \text{m}$$

SOURCE MICRO-ROUTE

EXAMPLE VALUES IN THE SOURCE

If $\Delta v = 20 \text{ m/s}$ and $a = 4 \text{ m/s}^2$, then $t = 20/4$. If the average velocity is 15 m/s , then displacement is $15t$. The point is the route: change/rate gives time; average velocity with time gives displacement.

PHYSICAL READING

Acceleration tells how fast velocity changes. Dividing the total change by that rate gives the duration. Average velocity then converts that duration into displacement.

M-ST06C

Worked example + misconception repair

Use change, rate, average - but do not overbuild the problem

WORKED EXAMPLE

QUESTION

During one interval, a particle's velocity increases by 24 m/s under constant acceleration 6 m/s². Its average velocity over that same interval is 18 m/s. Find the displacement.

FIRST MOVE

Find the duration from velocity change and acceleration. Do not invent initial and final velocities.

SOLUTION

$$t = 24/6 = 4 \text{ s}$$

$$s = 18 \times 4 = 72 \text{ m}$$

CHECK

72 m in 4 s corresponds to 18 m/s average velocity, exactly as stated.

MISCONCEPTION CLINIC

WRONG IDEA

"Every constant-acceleration problem must begin by filling u, v, a, t, s."

WHY IT FEELS SAFE

The UVATS strip is useful, but it is a tool, not a requirement. A familiar template can feel safer, yet extra unknowns increase algebra and opportunities for sign errors.

REPAIR

Ask for the shortest executable route: What is directly known? What is the target? Can a definition connect them without reconstructing unnecessary variables?

NOT AN ANTI-FORMULA RULE

Use standard kinematic equations when they are the clean route. The goal is model economy, not avoiding formulas.

M-ST06C

One-second data hides the same route

Decode the interval before recovering endpoint velocities

GIVEN OVER ONE SECOND

Displacement is 125 m and velocity increases by 50 m/s during the one-second interval $t \rightarrow t+1$ s.

READ THE DATA FIRST

INTERVAL

$$\Delta t = 1 \text{ s}$$

ACCELERATION

$$a = \Delta v / \Delta t = 50 \text{ m/s}^2$$

AVERAGE VELOCITY

$$125 \text{ m} / 1 \text{ s} = 125 \text{ m/s}$$

ONLY THEN RECOVER ENDPOINT VELOCITIES IF NEEDED

$$v - u = 50$$

$$(u + v) / 2 = 125$$

WHY THIS MATTERS

The definitions first reveal the interval's acceleration and average velocity. Reconstruct u and v only if a later question actually requires them.

DECISION STRIP

Known as change + average? → use definitions first. Need endpoints later? → reconstruct them after the interval is decoded. Different stages? → split the motion before using the shortcut.

M-ST06C

Guided practice

Keep the route compact, then test the model gate

FULL-SUPPORT MODEL

$$t = \Delta v / a$$

$$s = v_{\text{avg}} t$$

GUIDED QUESTION

A body gains 30 m/s of velocity under constant acceleration 5 m/s². Its average velocity in that interval is 20 m/s. Find the displacement.

HINT LADDER

H1: find time from change/rate.

H2: $t = 30/5$.

H3: multiply average velocity by that time.

WORK THE ROUTE

1. TIME FROM CHANGE/RATE

2. DISPLACEMENT FROM AVERAGE VELOCITY

3. UNIT / PHYSICAL CHECK

INDEPENDENT VARIANT

What if the average velocity given belongs to a different stage? The shortcut is invalid until the stages are separated.

MODEL GATE

All three pieces - Δv , a , and average velocity - must refer to the same interval.

M-ST07A

What g really means

Gravity is constant acceleration near Earth's surface - not a new formula chapter

PHYSICAL MEANING

NEAR EARTH'S SURFACE

Ignoring air resistance, a freely falling object's downward velocity changes by about 9.8 m/s every second.

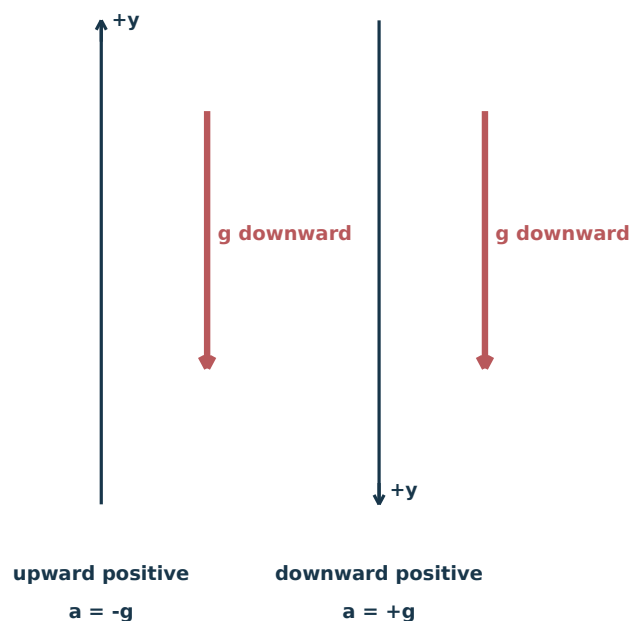
$$g \approx 9.8 \text{ m/s}^2$$

magnitude of the gravitational acceleration

MEMORY ANCHOR

g points downward. The sign is chosen by your axis: upward positive $\rightarrow a = -g$; downward positive $\rightarrow a = +g$.

DIRECTION IS PHYSICAL; SIGN IS CHOSEN



PHYSICAL READING

Each second, gravity changes velocity by about 9.8 m/s in the downward direction. The algebraic sign only encodes your chosen axis.

M-ST07A

Same physics, different signs

Coordinate systems change symbols, not the motion

UPWARD POSITIVE

$$a = -g$$

$$v = u - gt$$

INTERPRETATION

A falling object can have negative velocity and negative acceleration because both point downward when upward is chosen positive.

DOWNWARD POSITIVE

$$a = +g$$

$$v = u + gt$$

SAME MOTION

Nothing physical changed. Only the sign convention changed.

ONE EVENT, TWO DESCRIPTIONS

POSITIVE DIRECTION	up	down
GRAVITY DIRECTION	down	down
ACCELERATION SYMBOL	-g	+g
FALLING VELOCITY SIGN	negative	positive

M-ST07A

Gravity uses the same equation family

Choose the axis, then substitute $a = \pm g$ consistently

UPWARD-POSITIVE FORM

$$v = u - gt$$

$$y = ut - \frac{1}{2}gt^2$$

$$v^2 = u^2 - 2gy$$

NOT NEW FORMULAS

These are the same constant-acceleration relations already learned, with $a = -g$.

VERTICAL SIGN STRIP

1. Choose the positive direction.
2. Mark gravity downward.
3. Assign $a = +g$ or $-g$.
4. Translate u , v and y using the same axis.

MISCONCEPTION CLINIC

WRONG IDEA

"If $a = -g$, gravity is negative or weaker."

WHY IT HAPPENS

The minus sign is often read as physical size rather than direction.

CORRECT MODEL

Magnitude: $g \approx 9.8 \text{ m/s}^2$. Direction: downward.
Sign: determined by the chosen axis.

MODEL GATE

This school-level model assumes near-Earth motion and neglects air resistance unless told otherwise.

M-ST07A

Guided practice

Sign first, equation second

QUESTION

A stone is thrown upward. Choose upward as positive. Write the signs of u , a and the velocity v while the stone is still rising.

BUILD THE SIGN MODEL



YOUR SIGN TABLE

INITIAL VELOCITY u

sign = _____

ACCELERATION a

sign / form = _____

VELOCITY WHILE RISING v

sign = _____

HINT LADDER

H1: draw $+y$ upward.

H2: gravity arrow is downward.

H3: $u > 0$, $a = -g$, and $v > 0$ until the top.

REASON IN WORDS

Why does the sign of acceleration stay negative while the stone is rising, even though its velocity is positive?

M-ST07B

Free fall from rest

Velocity grows with t ; distance grows with t^2

CORE MODEL

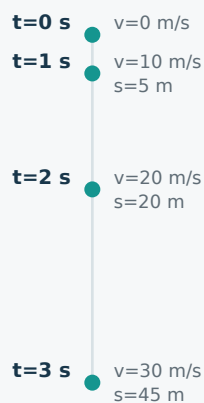
Choose downward as positive: $u = 0$ and $a = +g$

$$v = gt$$

$$s = \frac{1}{2}gt^2$$

Same constant-acceleration physics; free fall from rest only fixes $u = 0$ and $a = g$.

VISUAL BRIDGE ($g = 10 \text{ m/s}^2$)



Equal time steps: velocity rises evenly; distance gaps grow.

VELOCITY GROWS LINEARLY

Each second adds the same g to velocity. Equal time steps therefore create equal velocity increments.

DISTANCE GROWS QUADRATICALLY

Later seconds begin with larger speed, so cumulative displacement follows t^2 rather than t .

MEMORY ANCHOR

$v \propto t$, $s \propto t^2$.

Doubling fall time doubles speed but quadruples distance.

M-ST07B

Height, fall time, and a time-free example

Use the representation that matches the question

HEIGHT AND FALL TIME

$$h = \frac{1}{2}gt^2$$

$$t = \sqrt{(2h/g)}$$

time scales with the square root of height

SCALING

height $\times 4$

$$\sqrt{4} = 2$$

time $\times 2$

FIRST HALF VS SECOND HALF

The second half of the fall takes less time than the first half because the body reaches the midpoint already moving.

WORKED EXAMPLE

QUESTION

A ball is dropped from rest from 18 m. Using $g = 10 \text{ m/s}^2$, at what height above ground is $|v|$ numerically equal to $|g|$?

TRANSLATE

Numerically equal means speed = 10 m/s. Use the time-free relation $v^2 = 2gs$ for the distance fallen.

$$10^2 = 2(10)s$$

$$s = 5 \text{ m}$$

$$\text{height} = 18 - 5 = 13 \text{ m}$$

PHYSICAL CHECK

The ball begins from zero speed, so reaching 10 m/s after 5 m of fall is plausible for $g = 10 \text{ m/s}^2$.

M-ST07B

Multiple drops and the double-time trap

Same g does not mean same elapsed time

MULTIPLE DROPS: ONE GLOBAL INSTANT

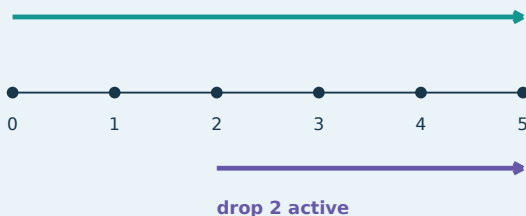
SHOWER-DROP STORY

Drops leave a nozzle at regular intervals. When the first drop reaches the floor, later drops have been falling for shorter times.

COMMON CLOCK

Older drop: longer local t Later drop: shorter local t

drop 1 active



$$s_i = \frac{1}{2}g t_i^2$$

same equation, different local elapsed time

CONNECT BACK

The common-clock idea from delayed-start motion supplies each drop's local elapsed time. Free-fall physics then supplies its position.

MISCONCEPTION CLINIC

WRONG IDEA

"If a body falls twice as long, it falls twice as far."

WHY IT FAILS

That would describe constant-speed motion. Free fall from rest is accelerated motion, so cumulative displacement follows t^2 .

COUNTEREXAMPLE

$$S(1) = 5 \text{ m}$$

$$S(2) = 20 \text{ m}$$

CORRECT MODEL

Doubling time makes distance four times as large. Velocity grows linearly; cumulative displacement grows quadratically.

M-ST07B

Guided practice: free fall from rest

Formula support is visible because this is full scaffold

QUESTION

A STONE IS DROPPED FROM REST AND FALLS FOR 3 S. TAKE $g = 10 \text{ m/s}^2$. FIND ITS SPEED AND DISTANCE FALLEN.

Start by choosing downward positive. A true drop from rest means $u = 0$.

MODEL

$$v = gt$$

$$s = \frac{1}{2}gt^2$$

H1: downward positive • H2: $u = 0$ • H3: substitute $t = 3 \text{ s}$ and $g = 10 \text{ m/s}^2$

YOUR WORKING

SPEED LANE

Known: $g = 10 \text{ m/s}^2$, $t = 3 \text{ s}$

Substitute into $v = gt$, then state the direction.

DISTANCE LANE

Known: $g = 10 \text{ m/s}^2$, $t = 3 \text{ s}$

Substitute into $s = \frac{1}{2}gt^2$, then check that the distance is positive.

CALCULATION

ANSWER + DIRECTION

CALCULATION

ANSWER + CHECK

NEXT IDEA

Free-fall “last distance” and “between two points” questions are interval problems: they require subtracting cumulative positions rather than restarting the fall.

M-ST07C

Last-distance and between-point fall problems

The last x metres are an interval, not a new drop from rest

CORE INTERVAL IDEA

MEMORY ANCHOR

LAST / BETWEEN -> subtract cumulative positions. The calculation interval may begin late in the fall, but the physical motion did not restart.

EXAMPLE: THE LAST 6 M TAKES 0.2 S



WRITE THE INTERVAL, THEN SUBSTITUTE FREE FALL

$$S(T) - S(T - 0.2) = 6$$

the body is already moving at $T - 0.2$

$$S(t) = \frac{1}{2}gt^2$$

cumulative free-fall position from release

$$\frac{1}{2}g[T^2 - (T - \Delta t)^2] = \Delta s$$

general last-interval form

WHY DIFFERENCE OF SQUARES APPEARS

The cumulative law is quadratic. Subtracting the two cumulative positions automatically carries the nonzero entry speed of the final interval without resetting $u = 0$.

M-ST07C

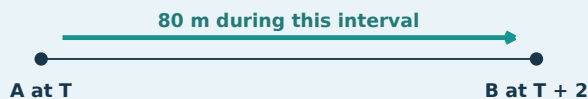
Between two points and spacing between drops

Different wording, same later-total minus earlier-total structure

BETWEEN POINT A AND POINT B

SITUATION

A falling body crosses point A, then 2 s later crosses point B, 80 m below A. The 80 m is an interval displacement.



$$S(T + 2) - S(T) = 80$$

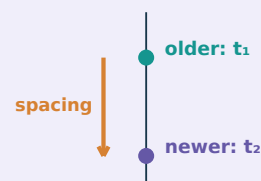
THEN FIND A

Once T is found, the distance of point A from the release point is S(T).

DROPLET SPACING AT ONE GLOBAL INSTANT

TWO DROPS OBSERVED TOGETHER

The older drop has been falling longer. Their vertical separation is the difference between two cumulative free-fall distances.



$$\text{spacing} = \frac{1}{2}g(t_1^2 - t_2^2)$$

CONNECT BACK

The common clock gives each drop its elapsed time. This interval model subtracts their cumulative fall distances.

RECOGNITION CUE

Words to watch: last x metres • between A and B • spacing between drops • during final Δt . One core move: later cumulative position - earlier cumulative position.

M-ST07C

Do not reset $u = 0$ mid-fall

A new calculation interval is not a physical restart

MISCONCEPTION CLINIC

WRONG IDEA

“For the last 0.2 s, use $u = 0$ and $t = 0.2$.”

WHY IT HAPPENS

Students can confuse the start of a calculation interval with a physical restart. But the object is already moving at the interval boundary.

SAFE ROUTE 1

Subtract cumulative positions: $S(\text{end}) - S(\text{start})$. This automatically includes the speed already acquired.

SAFE ROUTE 2

Alternatively, calculate the velocity at the interval start and carry it forward as the new initial velocity. Do not silently set it to zero.

GUIDED INTERVAL EQUATION

QUESTION

A stone is dropped from rest. It travels 70 m during a 2 s interval. Let the interval start T seconds after release. Write the equation needed to find T .

HINT LADDER

H1: the interval is $T \rightarrow T + 2$.

H2: use cumulative $S(t)$.

H3: later total - earlier total = 70.

$$S(T + 2) - S(T) = 70$$

write this before expanding $\frac{1}{2}gt^2$

YOUR SETUP

WRITE $S(T + 2)$

WRITE $S(T)$

FORM THE DIFFERENCE

NEXT IDEA

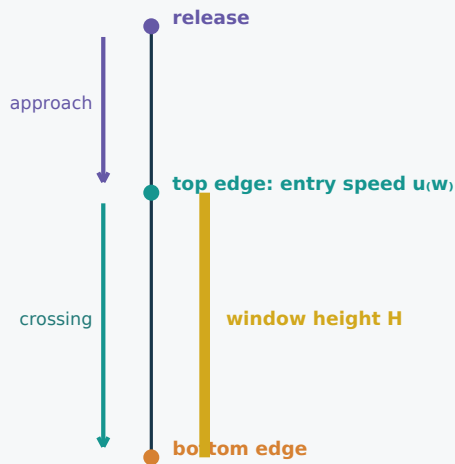
Window-crossing problems use the same no-reset principle: separate the approach stage from the local crossing stage and carry the entry speed forward.

M-ST07D

Window crossing: entry speed matters

Separate approach motion from the local window interval

TWO-STAGE MODEL



Do not reset $u = 0$ at the window unless the object is released there.

STAGE 1: REACH THE TOP EDGE

$$u(w)^2 = 2g h(\text{above})$$

release from rest above the window

ENTRY SPEED

A higher release point gives more fall distance before the window, so the object enters the window faster.

STAGE 2: CROSS THE WINDOW

$$H = u(w)\tau + \frac{1}{2}g\tau^2$$

same window height H ; local crossing time τ

COMPARE THE SAME H

If H and g are fixed, a larger entry speed $u(w)$ means a smaller positive crossing time τ .

MEMORY ANCHOR

HIGHER RELEASE → GREATER ENTRY SPEED → LESS WINDOW TIME. The whole journey may be longer, while the local time inside the same window is shorter.

M-ST07D

Why higher can mean faster through the window

Total journey time and local crossing time are different questions

MISCONCEPTION CLINIC

WRONG IDEA

“Dropped from higher means the whole trip is longer, so crossing the window must take longer too.”

WHY IT FEELS PLAUSIBLE

A greater release height does increase total fall time. But it also increases the speed at which the object arrives at the window.

CORRECT LOCAL COMPARISON

Inside the same window, both cases cover the same distance H under the same g . The case with larger entry speed finishes that local interval sooner.

WORKED EXAMPLE

QUESTION

A 5 m tall window is crossed by a falling ball. Case A: released from rest at the top edge. Case B: released 15 m above the top edge. Take $g = 10 \text{ m/s}^2$. Which case has the shorter crossing time?

$$\text{Case A: } 5 = 5\tau^2 \rightarrow \tau = 1 \text{ s}$$

$$\text{Case B: } u(w) = \sqrt{300}$$

NO EXACT T NEEDED

Because $u(w) > 0$ while H and g are unchanged, the positive crossing time in Case B must be less than 1 s.

PHYSICAL READING

The hidden variable is the speed at entry to the window, not the total distance fallen since release.

M-ST07D

Approach → entry → crossing

Carry the entry speed forward; do not reset the motion

THREE-STEP HELPER

APPROACH

1

Use the release height to find the speed at the top edge.

ENTRY

2

Carry that speed into the window stage.

CROSS

3

Use $H = u_{(W)}\tau + \frac{1}{2}g\tau^2$ for the local interval.

MODEL GATE

This school-level model assumes constant downward g and negligible air resistance. The object enters the window with the speed acquired above it.

GUIDED SETUP

QUESTION

A ball is dropped 20 m above the top of a 4 m window. Take $g = 10 \text{ m/s}^2$. Write the two equations needed to determine its window-crossing time.

HINT LADDER

H1: find entry speed after 20 m.

H2: use $v^2 = 2gh$.

H3: use $4 = u_{(W)}\tau + 5\tau^2$.

YOUR EQUATIONS

ENTRY-SPEED RELATION

WINDOW-CROSSING RELATION

NEXT IDEA

A vertical throw has a turning instant where $v = 0$, but gravity still acts. That distinction becomes the anchor for the next subtopic.

M-ST08A

Highest point: $v = 0$, $a = -g$

The object pauses in velocity, not in acceleration

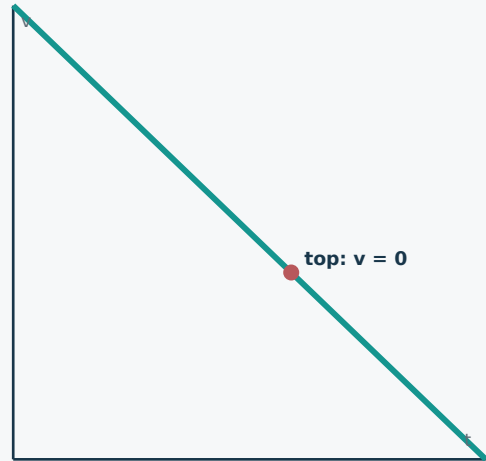
USE UPWARD AS POSITIVE

$$a = -g$$

$$v = u - gt$$

gravity subtracts g from upward velocity each second**AT THE TOP**

$v = 0$ is the turning-instant condition. The acceleration remains $-g$, which is exactly what makes velocity become negative immediately afterward.

Straight-line slope = $-g$ through the zero-velocity crossing.**GRAPH CHECK**

The v - t line crosses zero with a nonzero negative slope. That slope is acceleration.

MEMORY ANCHOR

TOP $\rightarrow v = 0$, but gravity still acts. Velocity describes current motion; acceleration describes how velocity is changing.

M-ST08A

 $v = 0$ does not imply $a = 0$

Velocity and acceleration answer different questions

MISCONCEPTION CLINIC**WRONG IDEA**

“At the highest point the ball stops, so acceleration must also be zero.”

COUNTEREXAMPLE

At the top, $v = 0$ but $a = -g$. A moment later the velocity is downward, so something nonzero had to change it.

CORRECT MODEL

Keep $a = -g$ continuously through the turning point. Use $v = 0$ only as the condition for the top itself.

WORKED EXAMPLE**QUESTION**

A ball is thrown vertically upward at 150 m/s. Take $g = 10 \text{ m/s}^2$. Compare its velocities after 3 s and 5 s.

$$v(3) = 150 - 10 \cdot 3 = 120 \text{ m/s}$$

$$v(5) = 150 - 10 \cdot 5 = 100 \text{ m/s}$$

PHYSICAL CHECK

Both velocities are still upward because the time to the top is 15 s.

WHEN NOT TO SET $v = 0$

Only use $v = 0$ when the question refers to the highest point or turning instant, not merely any later time.

M-ST08A

Guided turning-point reasoning

Sign first, then use the top condition

QUESTION

QUESTION

A stone is thrown vertically upward with $u = 30 \text{ m/s}$. Take $g = 10 \text{ m/s}^2$. Find the time when its velocity first becomes zero and state its acceleration at that instant.

MODEL STRIP

top: $v = 0$

$0 = 30 - 10t$

$a = -10 \text{ m/s}^2$

H1: top means $v = 0$ • H2: acceleration is still $-g$ • H3: solve the velocity equation

YOUR WORKING

SOLVE FOR T

STATE ACCELERATION AT THE TOP

EXPLAIN WHY A IS NOT ZERO

NEXT IDEA

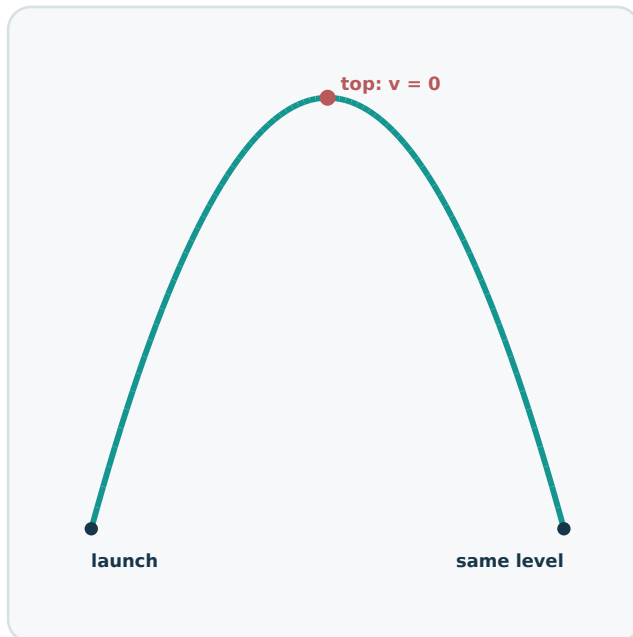
Time to top, maximum height, and same-level flight time are three results built from this same turning-point condition $v = 0$.

M-ST08B

Time to top, maximum height, flight time

Three results built from one turning-point condition

START FROM $v = 0$ AT THE TOP



BUILD, DO NOT MEMORISE SEPARATELY

$$0 = u - gt_{(up)}$$

$$t_{(up)} = u/g$$

$$0 = u^2 - 2gH$$

$$H = u^2/(2g)$$

$$T = 2u/g$$

only for return to the launch height

MODEL GATE

Same-level flight time uses symmetry: time down = time up. Do not use $T = 2u/g$ when the landing height differs from the launch height.

MEMORY ANCHOR

TOP CONDITION $\rightarrow v = 0$. Use it once for time, once for height. The flight-time result adds the same-level return condition.

M-ST08B

Scaling and hidden timing language

Read the powers before recalculating

SCALING AT THE SAME G

CHANGE LAUNCH SPEED U

$$H \propto u^2$$

$$T \propto u$$

Double $u \rightarrow$ maximum height becomes $4\times$
while same-level flight time becomes $2\times$.

CHANGE G

$$H \propto 1/g$$

$$t_{\text{(up)}} \propto 1/g$$

Before computing, identify whether u , g , or the launch/landing height changed.

COMPARE PARAMETERS

Do not mix scaling rules. The proportionality you use depends on which quantities are held fixed.

WORKED EXAMPLE: JUGGLER RELEASE RATE

QUESTION

A juggler throws the next ball exactly when the previous ball reaches its highest point. If the juggler throws n balls each second, find the maximum height.

TRANSLATE THE RATE

Time between throws = $1/n$. That equals $t_{\text{(up)}}$.

$$u/g = 1/n \rightarrow u = g/n$$

$$H = g/(2n^2)$$

WHY THIS WORKS

The problem hides the upward flight time inside a throwing frequency.

M-ST08B

Same-level flight time has a condition

Symmetry is powerful, but only when the endpoints match

MISCONCEPTION CLINIC

WRONG IDEA

"Total flight time is always $2u/g$."

COUNTEREXAMPLE

Throw from the top of a tower and land on the ground: the downward journey extends below the launch level, so the total time is longer.

CORRECT MODEL

If final height = initial height, use symmetry. Otherwise solve the position equation for the actual landing level.

GUIDED

QUESTION

A ball is thrown upward at 40 m/s from ground. Take $g = 10$. Find time to top, maximum height, and total same-level flight time.

HINT LADDER

H1: top gives $v = 0$.

H2: $t_{\text{up}} = u/g$.

H3: use $H = u^2/(2g)$, then double t_{up} .

YOUR WORKING

TIME TO TOP

MAXIMUM HEIGHT

FLIGHT TIME + CONDITION

NEXT IDEA

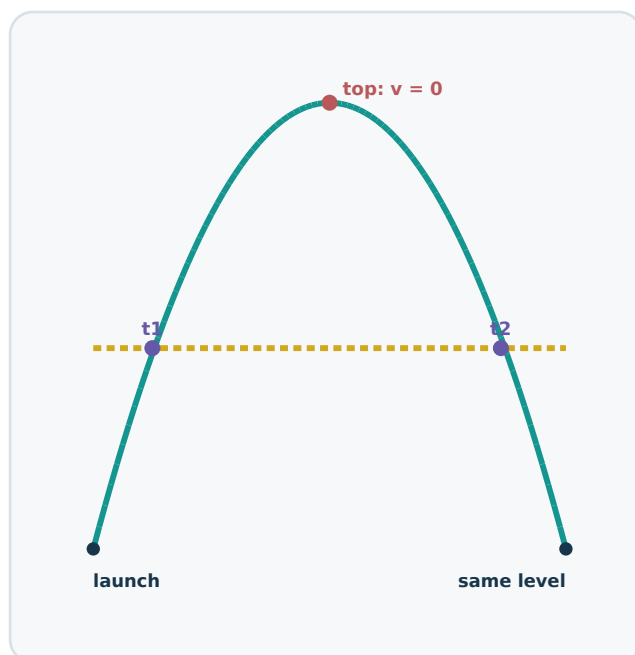
One height below the maximum is crossed twice. The two passage times become the two roots of the same position equation.

M-ST08C

Same height twice

One height, two passage times: upward and downward

PHYSICAL PICTURE



TWO PASSAGES

The earlier time t_1 is on the way up; the later time t_2 is on the way down.

ONE QUADRATIC

$$y = ut - \frac{1}{2}gt^2$$

$$\frac{1}{2}gt^2 - ut + y = 0$$

fix y ; solve for t

MEMORY ANCHOR

ONE HEIGHT → TWO ROOTS. The same height below the maximum can occur at an upward and a downward passage.

WHY A QUADRATIC APPEARS

The vertical position is quadratic in time. Fixing the height turns the motion equation into a quadratic whose two physical roots are the two passage times.

M-ST08C

Sum of the times and opposite velocity signs

Quadratic algebra and flight symmetry tell the same story

SUM OF ROOTS

$$\text{For } At^2 + Bt + C = 0$$

$$t_1 + t_2 = -B/A$$

$$t_1 + t_2 = 2u/g$$

for $y = ut - \frac{1}{2}gt^2$

SAME HEIGHT, SAME SPEED MAGNITUDE

$$v^2 = u^2 - 2gy$$

$$v(\text{up}) = +\sqrt{(u^2 - 2gy)}$$

$$v(\text{down}) = -\sqrt{(u^2 - 2gy)}$$

MEANING

The sum equals the total same-level flight time. Algebra and symmetry agree.

CHECK THE QUESTION WORD

If it asks speed, use magnitude. If it asks velocity, include direction/sign.

MISCONCEPTION REPAIR

Same height does not mean same velocity. Under the ideal model the speed magnitudes match, but the upward passage has positive velocity and the downward passage negative velocity when upward is positive.

M-ST08C

Worked idea and guided proof

Use structure before solving the quadratic completely

WORKED IDEA

HEIGHT $h/3$

A ball reaches maximum height h . At height $h/3$ it is seen once going up and once coming down. The two times are roots of one quadratic.

FIRST MOVE

Use $H = u^2/(2g)$ to express u^2 in terms of gh , then solve the height equation or use root relations.

CONCEPT TO KEEP

A repeated height is not two different motions. It is one trajectory intersecting the same horizontal level twice.

GUIDED TRANSFER

QUESTION

A ball is thrown upward at u . It crosses height y at t_1 and t_2 . Show without solving the quadratic completely that $t_1 + t_2 = 2u/g$.

HINT LADDER

H1: rearrange the position relation into a quadratic in t .

H2: identify A and B.

H3: use the root sum $-B/A$.

YOUR PROOF

QUADRATIC FORM

IDENTIFY A AND B

APPLY ROOT SUM

NEXT IDEA

M-ST08D: a fraction of launch speed maps to a different fraction of maximum height because the no-time relation contains v^2 .

M-ST08D

Speed fraction vs height fraction

Height depends on v^2 , so square the speed fraction first

$$v^2 = u^2 - 2gy$$

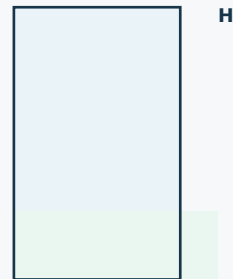
same no-time relation; now read it as a fraction map

SUPPOSE $v = ku$

$$k^2 u^2 = u^2 - 2gy$$

$$y/H = 1 - k^2$$

$$\text{remaining}/H = k^2$$



$$v = 0.5u$$

$$\text{reached} = 1 - k^2$$

$$\text{remaining} = k^2$$

MEMORY ANCHOR

SQUARE THE SPEED FRACTION FIRST. If $v = ku$, the fraction of maximum height already reached is $1 - k^2$; the fraction still remaining is k^2 .

M-ST08D

Linear fractions fail

The law is quadratic in speed

MISCONCEPTION CLINIC

WRONG IDEA

“Speed has fallen by half, so half the maximum height remains.”

WHY IT FAILS

The no-time relation contains v^2 . The relevant fraction is the square of v/u , not v/u itself.

CORRECT ROUTE

1. Form $k = v/u$.
2. Square it.
3. Reached fraction = $1 - k^2$.
4. Remaining fraction = k^2 .

COUNTEREXAMPLES MAKE IT VISIBLE

$$v = u/2 \rightarrow y = 3H/4$$

$$\text{remaining} = H/4$$

$$v = 0.8u \rightarrow y/H = 0.36$$

CONCEPT CHECK

Half the speed does not mean half the height. A speed fraction becomes a height fraction only after squaring.

M-ST08D

Use fractions before numbers

Worked example + guided practice

WORKED

QUESTION

A ball is thrown upward. At some point its speed is 60% of launch speed. What fraction of maximum height has it reached?

$$y/H = 1 - 0.6^2 = 0.64$$

ANSWER

64% of maximum height.

GUIDED

QUESTION

When the speed has fallen to one-third of u , what fraction of H remains above the body?

HINT LADDER

H1: $k = 1/3$.

H2: remaining fraction = k^2 .

H3: answer = $1/9$.

YOUR WORKING FORM K

SQUARE IT

STATE REMAINING FRACTION

NEXT

M-ST08E: keep the same equation and change only g .

M-ST08E

Earth vs Moon: change only g

Same governing equation; isolate the parameter that changed

SAME LAUNCH SPEED

g

g/6

H × 1

H × 6

EARTH

MOON

$$H = u^2/(2g)$$

$$\text{same } u \rightarrow H \propto 1/g$$

EARTH VS MOON EXAMPLE

If $g_{\text{Moon}} \approx g_{\text{Earth}}/6$ and u is the same, then $H_{\text{Moon}} \approx 6H_{\text{Earth}}$.

SOURCE CAUTION

Use the 1/6 numerical ratio only when it is supplied or already known. Do not invent the Moon ratio merely because the context changed.

M-ST08E

Reverse the question

Same height, different required launch speed

SOLVE THE SAME LAW FOR U

$$u = \sqrt{(2gH)}$$

$$u_{(M)}/u_{(E)} = \sqrt{(g_{(M)}/g_{(E)})}$$

$$g_{(M)} = g_{(E)}/6 \rightarrow \text{ratio} = 1/\sqrt{6}$$

PHYSICAL READING

For the same target height, smaller g requires a smaller launch speed.

MISCONCEPTION CLINIC

WRONG IDEA

"Earth and Moon need different motion formulas."

CORRECT MODEL

SAME LAW → CHANGE ONE PARAMETER. Write the same equation twice, label the environment, form a ratio, then insert numbers.

PARAMETER DISCIPLINE

Identify first what is held fixed: u, H, or g. The ratio changes depending on that choice.

M-ST08E

Parameter comparison

Guided: change the environment, not the law

QUESTION

A ball launched upward with the same speed reaches 20 m on Earth. If lunar g is one-sixth of Earth's, how high would the school-level model predict on the Moon?

$$\text{same } u \rightarrow H \propto 1/g$$

HINT LADDER

H1: same u .

H2: use $H \propto 1/g$.

H3: one-sixth $g \rightarrow$ six times H .

ASSUMPTION CHECK

The numerical Moon/Earth g ratio is explicitly supplied here. The governing constant-acceleration law is unchanged.

YOUR CALCULATION

SET RATIO

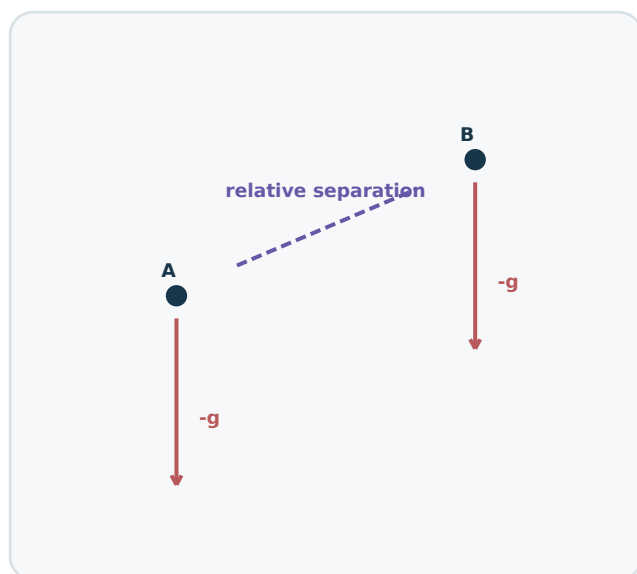
CALCULATE HEIGHT

STATE MODEL ASSUMPTION

M-ST09A

Relative acceleration: common gravity cancels

Gravity still acts on both bodies; only the separation equation loses it



SUBTRACT THE ACCELERATIONS

$$x_{(A/B)} = x_{(A)} - x_{(B)}$$

$$a_{(A/B)} = a_{(A)} - a_{(B)}$$

$$(-g) - (-g) = 0$$

IMPORTANT

Gravity did not vanish physically. It affects both bodies equally, so it cancels only when describing their separation.

WHAT FOLLOWS

Once both bodies are freely falling under the same g , their relative velocity stays constant because relative acceleration is zero.

M-ST09A

Delayed launch: common clock + relative motion

Two ideas combine after the second launch

AT THE SECOND LAUNCH

Ball 1 active

Ball 2 starts



STEP 1: SNAPSHOT BALL 1

At the instant Ball 2 is launched, find Ball 1's position and velocity. This is the handoff from the common clock.

STEP 2: SWITCH TO RELATIVE MOTION

After both are in flight under the same g , $a_{\text{rel}} = 0$. Use constant relative velocity to close the existing separation.

CONNECTION

M-ST06A supplies the delayed-start clock. M-ST09A supplies the relative-motion simplification. Do not use $a_{\text{rel}} = 0$ before both bodies share the same acceleration.

M-ST09A

Relative is not absolute

Misconception repair + guided recognition

MISCONCEPTION CLINIC

WRONG IDEA

“Relative acceleration is zero, so gravity is not acting.”

CORRECT MODEL

Absolute acceleration: $-g$ for each body. Relative acceleration: 0 between them. Ground-frame velocities still change; their difference stays constant.

GUIDED

QUESTION

Ball 1 is projected upward at 50 m/s. Two seconds later Ball 2 is projected upward from the same point at 50 m/s. Take $g = 10$. Find when Ball 2 meets Ball 1.

HINT LADDER

- H1: at $t = 2$ s, find Ball 1’s position and speed.
- H2: after $t = 2$ s, $a_{\text{rel}} = 0$.
- H3: relative speed is constant; close the separation.

YOUR WORKING

BALL 1 SNAPSHOT

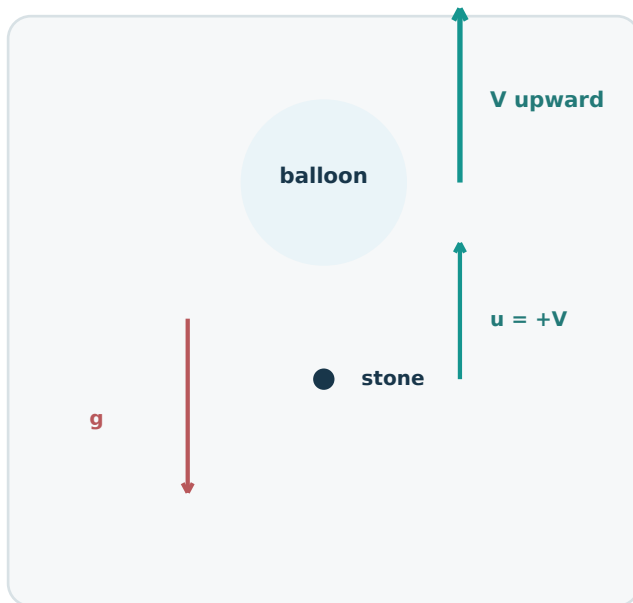
RELATIVE SPEED

MEETING TIME

M-ST09B

Dropped from a moving platform

"Dropped" means no extra throw relative to the carrier



REFERENCE FRAME FIRST

$$v_{(\text{stone/balloon})} = 0$$

RELATIVE TO BALLOON

No extra throw: the stone has zero release velocity relative to the balloon.

$$u_{(\text{stone,ground})} = V_{(\text{balloon})}$$

IF BALLOON RISES

With upward positive and carrier speed +V, the stone begins free flight with $u = +V$ and $a = -g$.

MEMORY ANCHOR

RELEASED FROM MOVING CARRIER → INHERIT CARRIER VELOCITY.

M-ST09B

The stone first keeps going up

Inertia before reversal

GROUND-FRAME STORY

IMMEDIATELY AFTER RELEASE

The stone is already moving upward. Gravity slows that upward motion, brings v to zero, then makes it fall.

$$y = y_0 + Vt - \frac{1}{2}gt^2$$

$$v = V - gt$$

DO NOT RESET $u = 0$

The free-flight stage begins with the carrier's instantaneous velocity.

MISCONCEPTION CLINIC

WRONG IDEA

"Dropped means $u = 0$."

WHY THAT CAN FAIL

It is true only when the releasing hand/platform is at rest in the chosen frame. A rising carrier gives the released object upward ground velocity.

FRAME → CARRIER → OBJECT

1. Choose observer/frame.
2. Read carrier velocity in that frame.
3. Add any release velocity relative to carrier. For a simple drop, that relative term is zero.

M-ST09B

Moving-carrier release

Guided: write the inherited initial condition first

QUESTION

A balloon rises at 10 m/s and releases a stone at height 75 m. Take $g = 10$. Write the stone's ground-frame initial velocity and the equation for its height after t seconds.

HINT LADDER

H1: use the ground frame.

H2: inherit $u = +10$ m/s.

H3: $y = 75 + 10t - 5t^2$.

$$u = +10 \text{ m/s}$$

$$y = 75 + 10t - 5t^2$$

EXPLAIN THE FRAME CHOICE

WHY u IS NOT ZERO

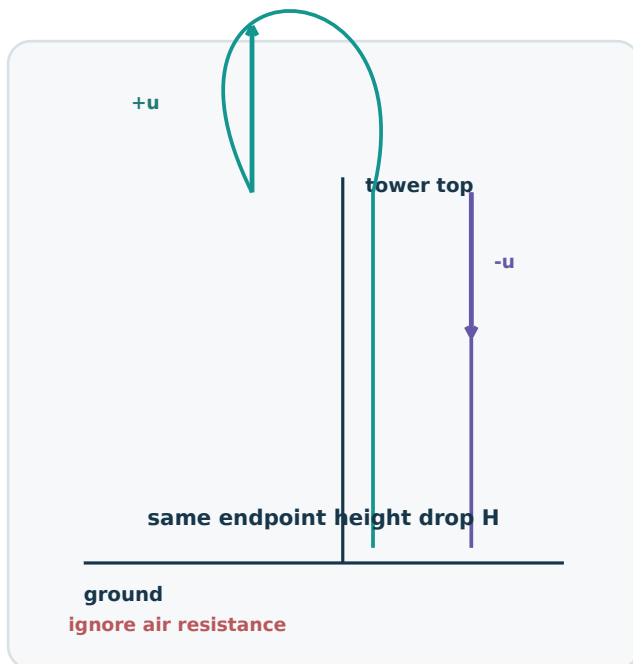
SIGN CONVENTION

PHYSICAL STORY AFTER RELEASE

M-ST09C

Upward vs downward launch: same impact speed

Initial direction changes the journey, not the ideal final speed magnitude



SAME START LEVEL + SAME $|u|$ + SAME LOWER ENDPOINT

$$v^2 = u^2 + 2gH$$

WHY THE SIGN DISAPPEARS

Whether launch velocity is $+u$ or $-u$, squaring gives the same u^2 . The same vertical drop H then gives the same final speed magnitude.

WHAT CHANGES

Path history and travel time differ. The upward-thrown ball rises first, returns, then follows the downward journey.

MEMORY ANCHOR

SAME $|u|$ + SAME HEIGHT DROP $\rightarrow$ SAME final speed magnitude. This is an endpoint result for the ideal constant- g , no-air-resistance model.

M-ST09C

Longer path does not mean greater final speed

Net height change controls v^2 in this model

MISCONCEPTION CLINIC

WRONG IDEA

"The upward-thrown ball travels farther, so it must hit faster."

WHY IT FEELS PLAUSIBLE

Distance travelled can matter when energy is dissipated. Here air resistance is neglected, so total path length is not the controlling variable.

CORRECT MODEL

$$v = \sqrt{u^2 + 2gH}$$

ENDPOINT TEST

Compare initial speed magnitude and net vertical displacement to the common final level. Do not substitute total path length for H .

MODEL GATE

If air resistance is important, the two paths can dissipate different energy and the impact-speed equality need not hold.

SPEED VS VELOCITY

The source claim is about equal final speed magnitudes. Velocity signs depend on the chosen axis and the fact that both are moving downward at impact.

M-ST09C

Compare magnitude, not travel time

Guided comparison from the same tower top

QUESTION

From the same tower top, one ball is thrown upward at 12 m/s and another downward at 12 m/s. Ignore air resistance. Compare their speeds at the ground.

$$\text{for each ball: } v^2 = u^2 + 2gH$$

HINT LADDER

H1: same $|u|$.H2: same net height H .H3: compare the two v^2 expressions before taking magnitudes.

PHYSICAL CHECK

One route is longer and takes longer, but both arrive at the same lower level with the same ideal speed magnitude.

YOUR WORKING

WRITE BOTH NO-TIME EQUATIONS

COMPARE SQUARED SPEEDS

STATE SPEED-MAGNITUDE CONCLUSION

M-ST09D

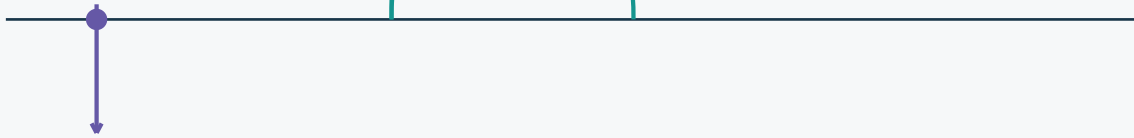
Time gap: upward vs downward launch

The upward launch contains one extra symmetric excursion

u/g up

u/g back

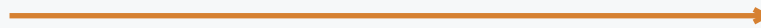
down launch



state: top, speed u downward

after $2u/g$: SAME STATE

same tower position + speed u downward



time shift $\Delta T = 2u/g$

STATE MATCHING

DOWNWARD-LAUNCHED BALL

Starts the downward journey immediately with speed magnitude u from the tower top.

UPWARD-LAUNCHED BALL

Spends u/g rising and u/g returning. At the tower top on the way down it matches the other ball's starting state.

$$\Delta T = 2u/g$$

M-ST09D

Paired times reveal the drop time

A geometric-mean consequence of the same tower setup

LET THE GROUND-HIT TIMES BE T_1 AND T_2

$$t_1 - t_2 = 2u/g$$

$$t_0 = \sqrt{(t_1 t_2)}$$

MEANING OF T_0

t_0 is the time to reach the ground if the ball is simply dropped from rest from the same tower.

MISCONCEPTION CLINIC

WRONG IDEA

"The upward launch only adds u/g ."

REPAIR

There are two extra symmetric legs: rise to the top and return to the launch level. Total added time is $2u/g$.

WHEN THE SHORTCUT APPLIES

Same tower, same $|u|$ used once upward and once downward, same g , same endpoint, ideal vertical motion.

M-ST09D

Tower time relations

Guided: use the paired-time relation before rebuilding the tower

QUESTION

A ball thrown upward from a tower takes 6 s to reach the ground. Thrown downward from the same point with the same speed, it takes 1.5 s. Find the time if simply dropped.

$$t_0 = \sqrt{t_1 t_2}$$

HINT LADDER

H1: the setup matches the paired-time relation.

H2: multiply the two hit times.

H3: take the positive square root.

$$t_0 = \sqrt{6 \times 1.5}$$

CHECK BEFORE USING IT

Do not use this shortcut if the two launch-speed magnitudes differ or the endpoints are not the same.

YOUR WORKING

MULTIPLY PAIRED TIMES

TAKE SQUARE ROOT

STATE THE MODEL CONDITIONS

M-ST10A

Position-time graph: slope is velocity

Read the axes first: graph height and graph slope mean different things

CORE READING

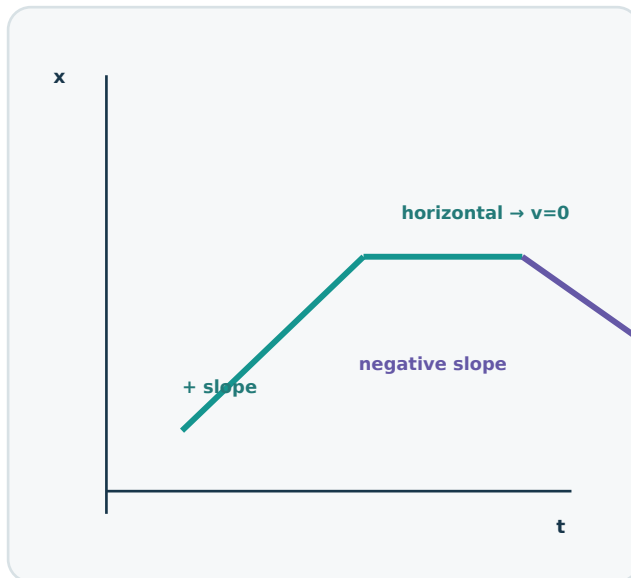
$$\text{slope} = \Delta x / \Delta t = \text{velocity}$$

GRAPH HEIGHT

The vertical coordinate tells position. A high horizontal segment can mean far from the origin but stationary.

SLOPE SIGN

Rising segment → positive velocity. Horizontal → $v=0$. Falling segment → negative velocity.



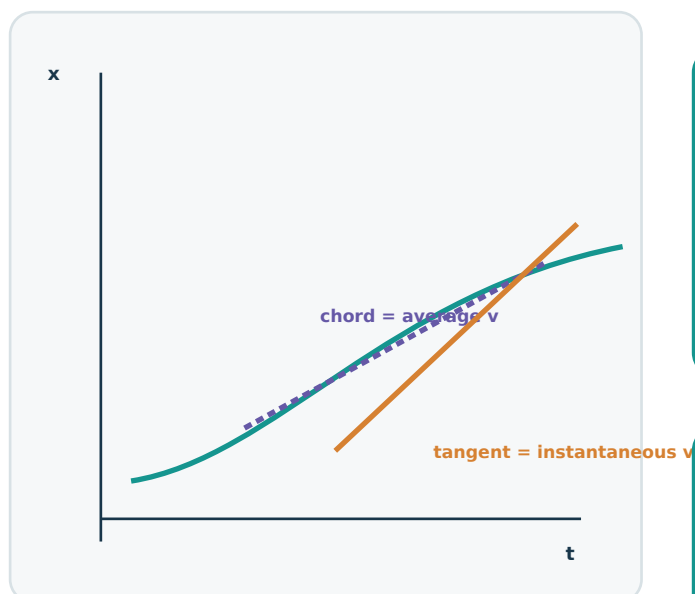
MEMORY ANCHOR

x-t graph height = position; x-t graph slope = velocity. Label axes before applying any graph rule.

M-ST10A

Chord for average, tangent for instantaneous

Mechanism + guided visual support



WHY IT WORKS

AVERAGE OVER AN INTERVAL

Use the chord between the two endpoint positions. Its slope is $\Delta x/\Delta t$ over that interval.

INSTANTANEOUS AT ONE TIME

Use the tangent slope at that instant. The tangent represents the limiting slope of shorter and shorter chords.

MISCONCEPTION CLINIC

WRONG IDEA

"Higher on an x-t graph means faster."

REPAIR

Speed/velocity comes from slope, not height. A high horizontal line means $v=0$.

GUIDED 1

A toy's x-t graph rises linearly, becomes horizontal for 2 s, then falls linearly. Interpret the velocity in each segment using slope sign.

M-ST10A

Independent transfer: same position after 12 s

No formula prompt: decide what the full-interval graph tells you

QUESTION

A cyclist's x - t graph has the same position at $t=0$ and $t=12$ s but changes in between. What is the average velocity over the full interval? What extra information would you need for distance?

FIRST MOVE

Draw or annotate the representation yourself before writing an equation. Use the two endpoint positions for average velocity; do not infer distance from endpoints alone.

WORK ZONE

ENDPOINT DISPLACEMENT

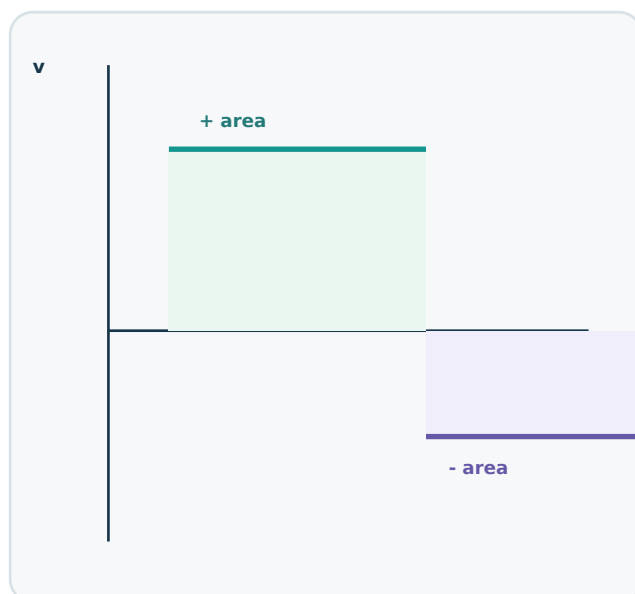
AVERAGE VELOCITY REASONING

WHAT INFORMATION IS MISSING FOR DISTANCE?

M-ST10B

Velocity-time graph: slope, signed area, distance

One graph carries acceleration and displacement information



READ TWO DIFFERENT GRAPH FEATURES

slope = $\Delta v / \Delta t$ = acceleration

signed area = displacement

absolute areas $\rightarrow$ distance

UNITS CHECK

Area has units $(\text{m/s}) \times \text{s} = \text{m}$. This is a physical check that the area quantity is a displacement.

MEMORY ANCHOR

v-t slope = acceleration; signed area = displacement; sum of absolute area magnitudes = distance when direction reverses.

M-ST10B

Use slope and area on the same page

Guided 1 + Guided 2 with the model visible

GUIDED 1: POSITIVE AND NEGATIVE VELOCITY



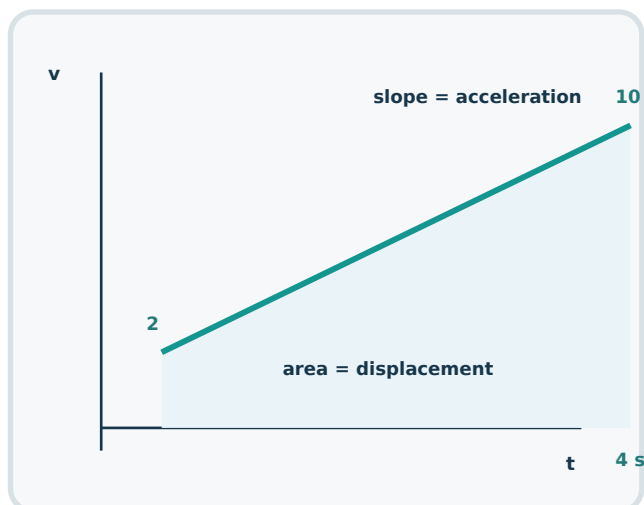
QUESTION

A v-t graph has +6 m/s for 4 s and -2 m/s for 3 s. Find displacement and distance.

HINT LADDER

H1: split above/below the axis.
 H2: signed area for displacement.
 H3: absolute areas for distance.

GUIDED 2: STRAIGHT RAMP



QUESTION

A straight v-t line rises from 2 to 10 m/s in 4 s. Find acceleration from slope and displacement from area.

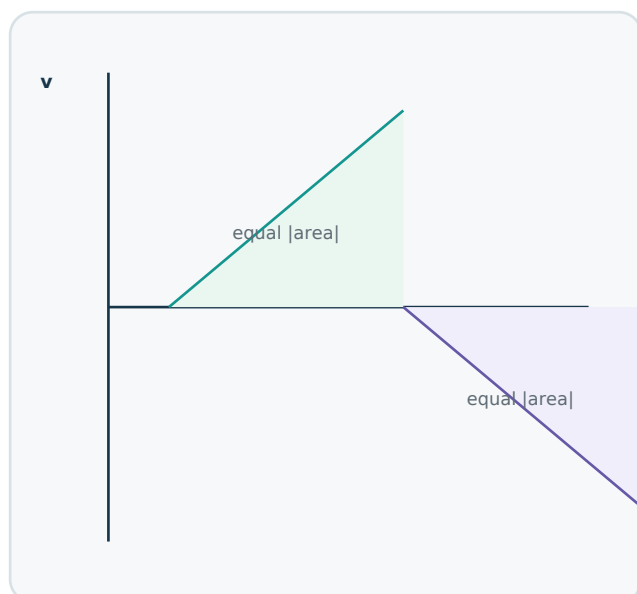
HINT LADDER

H1: $\text{slope} = (10 - 2) / 4$.
 H2: area is the trapezium under the line.
 H3: check units m/s^2 and m.

M-ST10B

Independent transfer: equal positive and negative areas

Net displacement can vanish while distance stays nonzero

**QUESTION**

A v-t graph has equal positive and negative signed areas. What are net displacement and average velocity? Can distance still be nonzero?

FIRST MOVE

Decide which question uses signed area and which uses absolute area. Do not convert the negative region into a positive displacement.

YOUR REASONING**NET SIGNED AREA**

AVERAGE VELOCITY OVER FULL INTERVAL

DISTANCE AND WHY IT CAN BE NONZERO

CONNECTION

For constant acceleration, splitting a v-t area into rectangle + triangle reproduces $s = ut + \frac{1}{2}at^2$.

M-ST10C

Vertical projection on a v-t graph

One straight line of slope $-g$ carries the whole free-flight story

$$v = u - gt$$

upward positive $\rightarrow$ slope = $-g$

READ THE LINE

Crossing $v=0$ marks the highest point. Below the time axis means negative velocity: motion downward relative to the chosen + upward axis.

COMMON TRAP

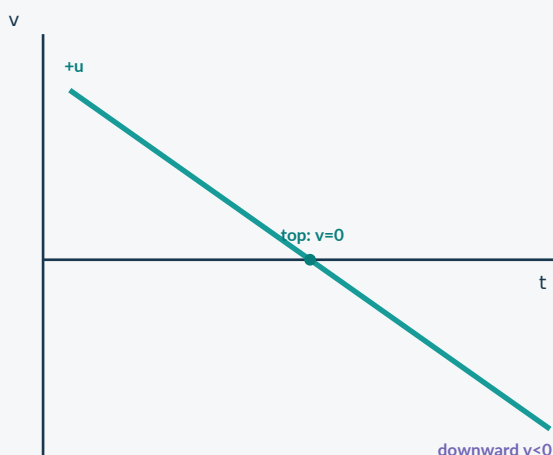
"Below the axis means negative speed." Speed is a magnitude and remains nonnegative; the sign belongs to velocity.

FIRST MOVE

Choose + upward first. During ideal free flight the line keeps the same slope $-g$ through the top; $v=0$ does not make $a=0$.

MEMORY ANCHOR

Vertical throw $\rightarrow$ one straight v-t line. Top = zero crossing; downward motion = $v < 0$; constant slope = $-g$.



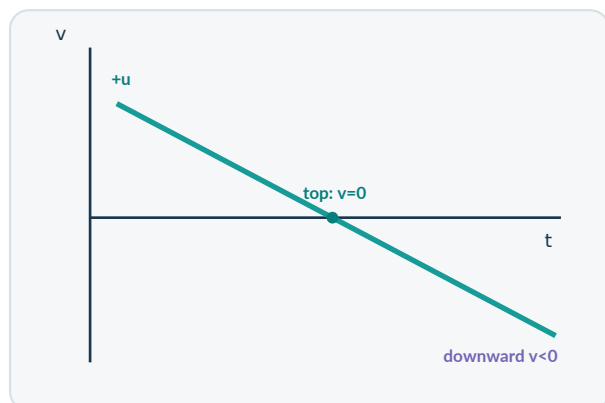
M-ST10C

Use the v-t line, then fade the support

Guided 1 gives the full graph; Guided 2 asks you to model a collision change

GUIDED 1 • FULL VISUAL

A ball is thrown upward at 30 m/s with $g=10 \text{ m/s}^2$. Sketch its v-t graph through 6 s and label the top.



MODEL → CALCULATE → CHECK

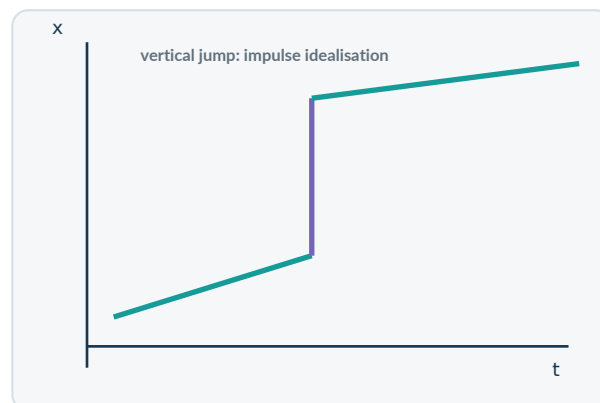
Slope = -10 m/s^2 . Set $v=0$: $0=30-10t \rightarrow$ top at 3 s.
Continue the same line to $v=-30 \text{ m/s}$ at 6 s.

HINT LADDER

H1: slope is $-g$.
H2: $v=0$ at $t=u/g$.
H3: continue the line below the axis after the top.

GUIDED 2 • FADED

A downward-moving body hits the ground and sticks. Describe the v-t graph immediately before and after impact.



Axes / sign choice:

Free-flight segment:

Impact / after sticking:

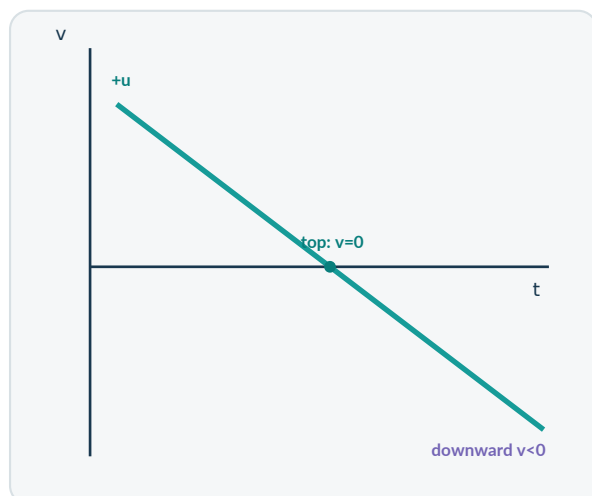
HINT LADDER

H1: choose sign.
H2: free-flight segment has fixed slope.
H3: a collision can create a velocity jump; after sticking, $v=0$.

M-ST10C

Independent transfer: zero crossing at 2.5 s

No formula prompt: infer the same-level flight time from graph symmetry



QUESTION

A vertical throw v - t line crosses zero at 2.5 s. Without knowing u , what is the total same-level flight time in the ideal model?

FIRST MOVE

Draw the v - t line yourself. Mark the zero crossing as the top, then use same-level symmetry rather than hunting for u .

WORK ZONE

YOUR GRAPH / ANNOTATION

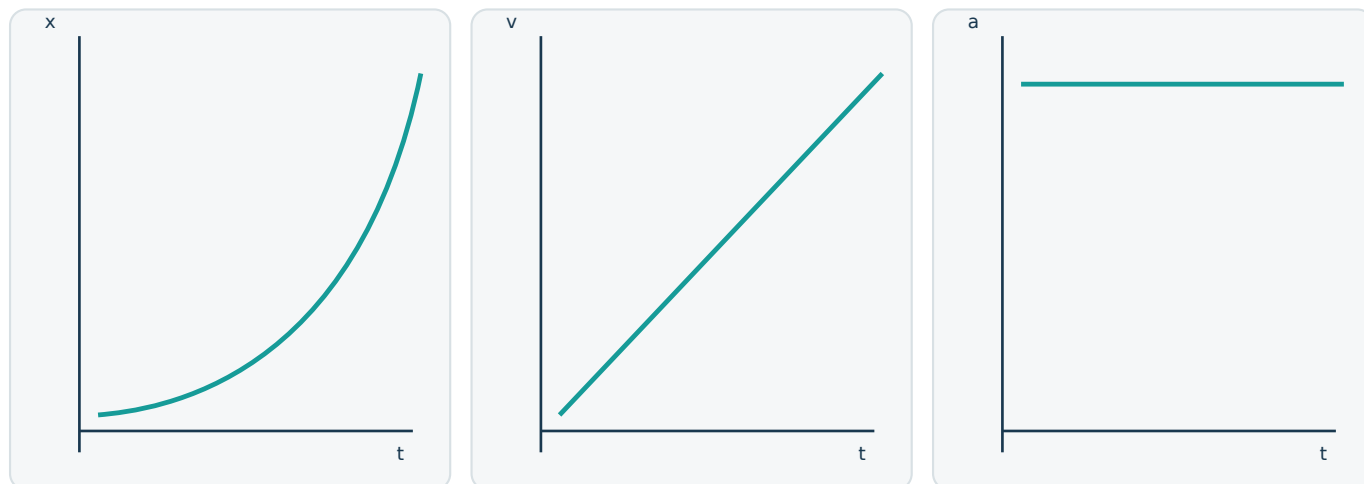
TIME-SYMMETRY REASONING

TOTAL FLIGHT TIME + CHECK

M-ST10D

Constant acceleration across x-t, v-t and a-t

One physical model, three linked graph shapes



a constant → v linear → x quadratic

$x = x_0 + ut + \frac{1}{2}at^2$ carries the t^2 curvature

WHY IT WORKS

Constant acceleration changes velocity steadily; accumulated position therefore acquires a t^2 term.

COMMON TRAP

“Constant acceleration gives a straight x-t graph.” Only v-t is straight under constant a. x-t is quadratic unless $a=0$.

MEMORY ANCHOR

Constant a → a-t horizontal, v-t straight, x-t quadratic. Curvature sign in x-t follows acceleration sign.

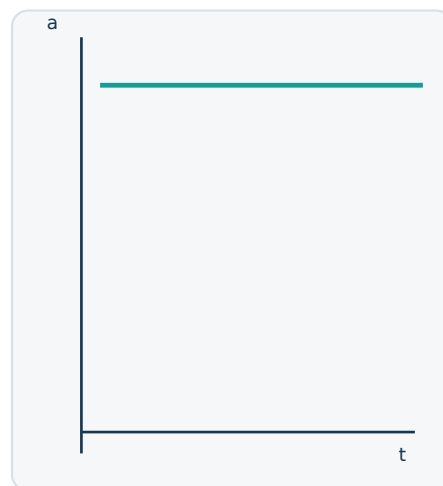
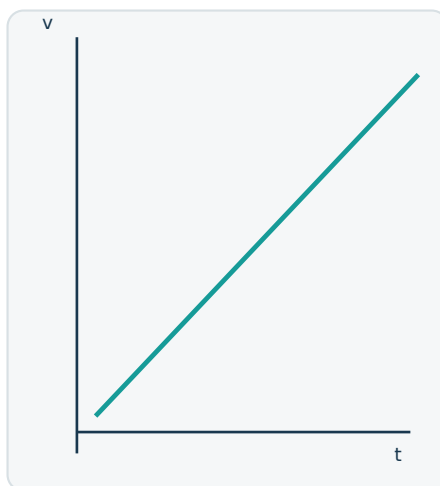
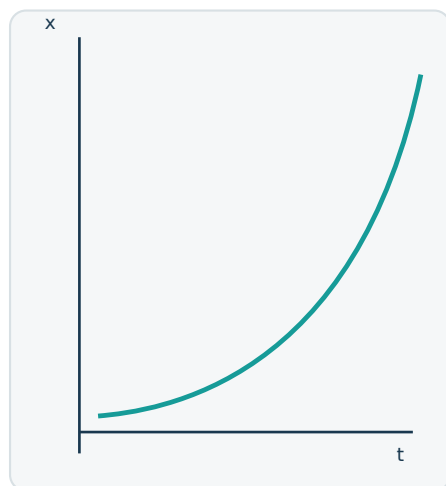
M-ST10D

Switch representations without changing the physics

Guided sketches make constant acceleration visible in three languages

GUIDED 1 • FULL VISUAL

A particle starts from rest with constant positive acceleration. Sketch a - t , v - t and x - t on one page.



HINT LADDER

H1: a is constant. H2: v changes linearly. H3: x accumulates quadratically.

GUIDED 2 • FADED

A body's x - t graph is a downward-opening parabola. What can you infer about the sign of acceleration?

REASONING GATE

Curvature indicates acceleration sign. Tangent slope is velocity. A turning point can have $v=0$ while curvature remains.

AXES / MODEL

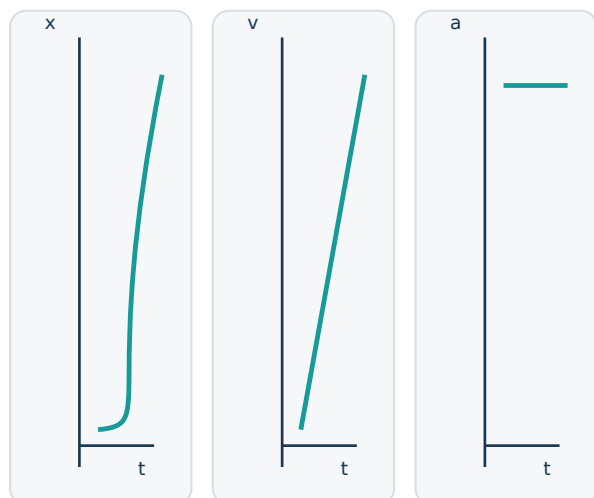
FIRST RELATION

PHYSICAL CHECK

M-ST10D

Independent transfer: identify constant acceleration

Choose the representation that shows the model most directly



QUESTION

Which representation makes constant acceleration easiest to see directly: x - t , v - t or a - t ? Explain how the other two must then look.

FIRST MOVE

Start from what each vertical axis means. Do not answer from memorised graph shapes alone.

WORK ZONE

DIRECT REPRESENTATION + WHY

HOW v - t MUST LOOK

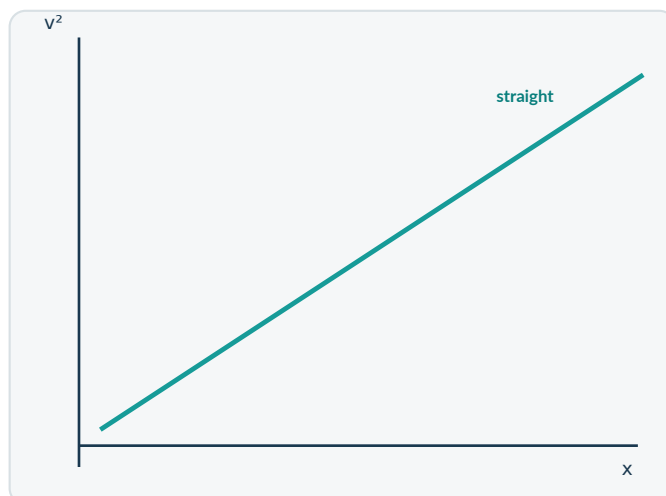
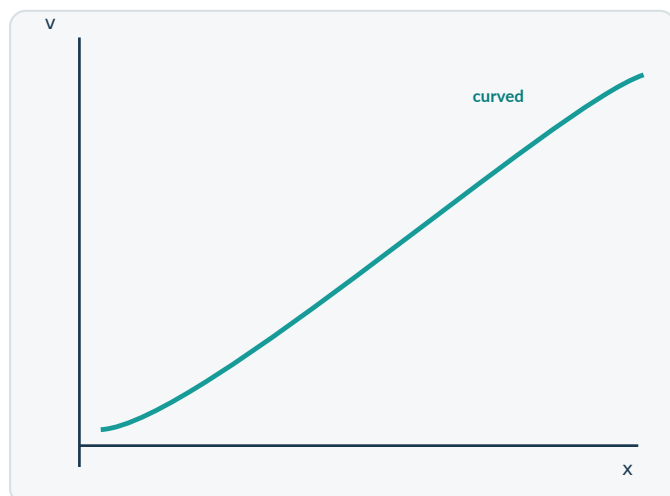
HOW x - t MUST LOOK

MODEL CHECK

M-ST10E

Velocity vs displacement: read the axes first

Constant acceleration is linear in v^2 versus x , not generally in v versus x



$$v^2 = u^2 + 2ax$$

straight line on a v^2 - x plot; slope = $2a$

AXIS DISCIPLINE

v - x and v^2 - x are different graphs. For free fall from rest, $v \propto \sqrt{x}$ while $v^2 \propto x$.

COMMON TRAP

"A velocity-displacement graph must be straight under constant acceleration." The straight relation belongs to v^2 versus x .

FOUNDATION EXTENSION

For advanced conversion, $a = v \cdot dv/dx$. Keep this labelled as an extension; it is not required to recognise the basic v^2 - x relation.

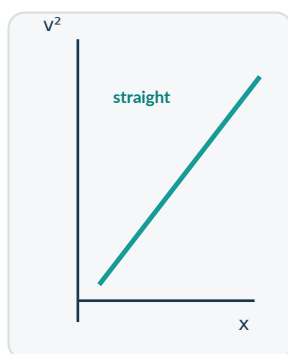
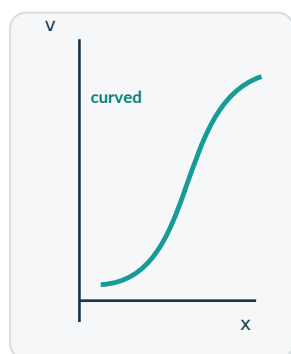
M-ST10E

Use the plotted variable before naming the curve

Guided 1 reads slope from v^2 -x; Guided 2 treats v -x as an extension

GUIDED 1 • FULL

For $v^2 = 4 + 6x$, what is the slope of the v^2 -x graph and what acceleration does it represent?



$$\text{slope} = 6 = 2a$$

$$a = 3 \text{ m/s}^2$$

HINT LADDER

H1: identify whether the vertical axis is v or v^2 .

H2: compare with $v^2 = u^2 + 2ax$.

H3: slope = $2a$.

GUIDED 2 • FADED

A v -x graph is straight with negative slope. Explain why the corresponding a -x graph need not have the same slope or shape.

EXTENSION MODEL

$a = v \cdot dv/dx$. Even if dv/dx is constant, multiplying by changing v makes a vary with x .

AXES / MODEL

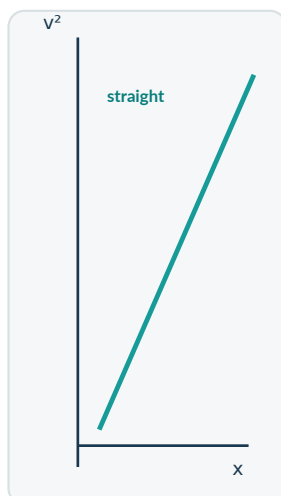
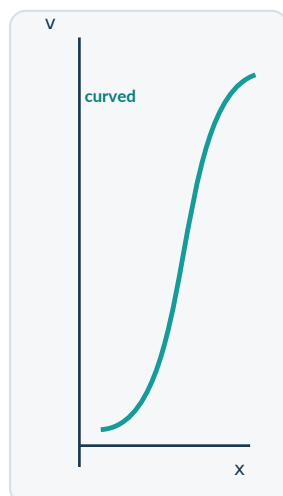
FIRST RELATION

PHYSICAL CHECK

M-ST10E

Independent transfer: straight v^2 -x graph

Read the plotted quantity before interpreting slope



QUESTION

A straight v^2 -x graph has slope -8 SI units. What is acceleration?
What happens to speed as x increases?

FIRST MOVE

Compare the graph with $v^2 = u^2 + 2ax$. The slope belongs to v^2 versus x , so it equals $2a$.

WORK ZONE

AXES + SLOPE MEANING

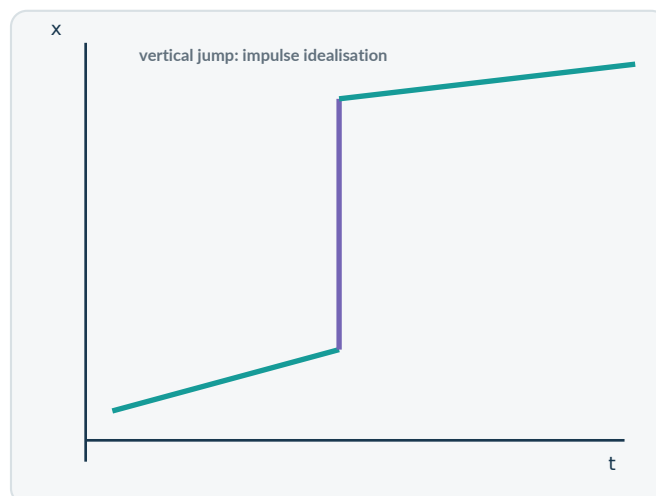
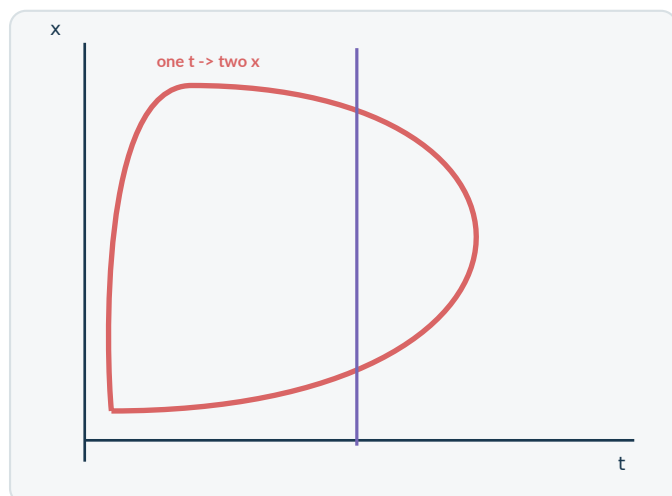
ACCELERATION

SPEED TREND + PHYSICAL CHECK

M-ST10F

Impossible and inconsistent motion graphs

A one-particle history must be single-valued in time and tell one consistent story



one particle + one instant $\rightarrow$ one position and one velocity

apply the vertical-line test to time-based graphs

WHY IT WORKS

A motion history is a function of time: one time input cannot map to two positions or two velocities for one particle in one frame.

DO NOT OVER-REJECT

A discontinuity or vertical v - t jump can idealise an impulse. A loop that gives two values at one time is fundamentally multi-valued.

PHYSICS NOTE

A finite velocity change in zero time corresponds to unbounded acceleration in the ideal limit. Across multiple graph types, also compare initial state, turning points and direction.

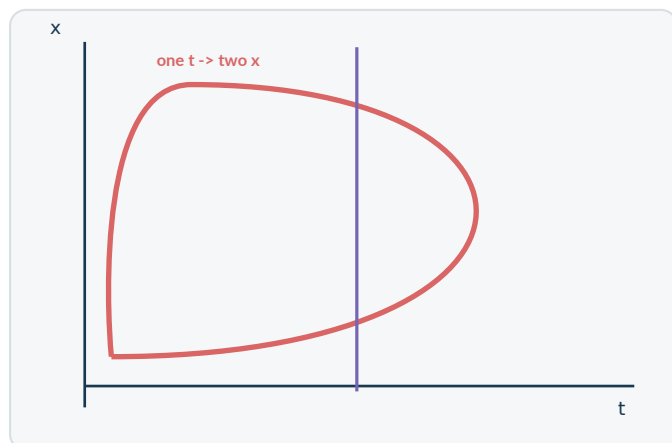
M-ST10F

Test validity first, then cross-check the story

Use the vertical-line test before comparing position, velocity and acceleration statements

GUIDED 1 • FULL

A proposed x-t graph is a closed loop. Explain exactly why it cannot be one particle's motion history.



MODEL → CHECK

Choose one time and draw a vertical line. More than one intersection means the graph gives more than one position at the same instant.

GUIDED 2 • FADED

Four graphs claim to describe the same motion. Write a checklist for converting each into position/velocity/acceleration statements before comparing.

AXES / MODEL

INITIAL STATE

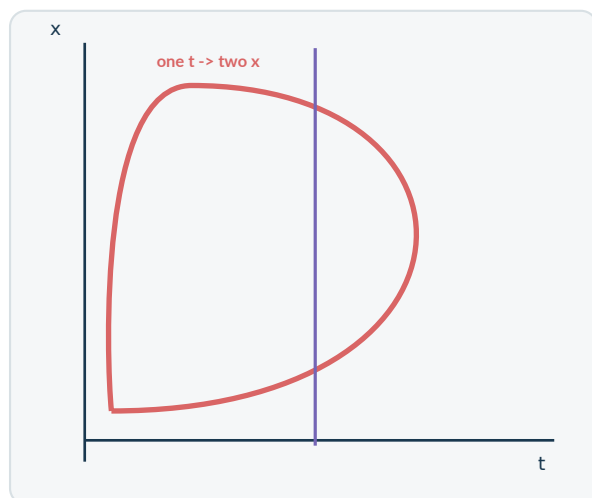
TURNING POINT / DIRECTION

CONSISTENCY CHECK

M-ST10F

Independent transfer: loop versus jump

Decide which graph feature is fundamentally impossible for a one-particle history



QUESTION

A v-t graph contains a loop. A second candidate contains one vertical jump but is otherwise single-valued. Which feature is fundamentally impossible for a one-particle history?

FIRST MOVE

Apply the vertical-line test first. Then distinguish a multi-valued history from an idealised impulse-like jump.

WORK ZONE

VERTICAL-LINE TEST

LOOP VERDICT + WHY

JUMP VERDICT + PHYSICS NOTE

M-ST11A

Average velocity numerically related to g

Foundation extension: translate the verbal number condition before solving



$$v = gt$$

free fall from rest

$$\bar{v} = gt/2$$

average of 0 and v under constant g

NUMERICAL CONDITION

If $\bar{v}$ has numerical value $g/2$, then $gt/2 = g/2 \rightarrow t=1\text{ s}$ and $v=g$.

FIRST MOVE

Write $v=gt$ and $\bar{v}=gt/2$ first. Only then translate “numerically equal to a multiple of g ” into an equation.

BOUNDARY / UNITS

A numerical comparison does not mean the quantities have the same dimensions. This shortcut assumes free fall from rest with constant g .

MEMORY ANCHOR

Translate the verbal number condition into an equation. General variant: $\bar{v} = (k g)/2$ numerically $\rightarrow t=k\text{ s}$ in the stated SI convention.

M-ST11A

Build the result instead of recalling a shortcut

Full support first; then generalise the same average-velocity structure

GUIDED 1 • FULL

A body falls freely from rest. Its average velocity over the elapsed time is numerically equal to g . Find the elapsed time.

GUIDED 2 • FADED

A freely falling body from rest has average velocity numerically equal to $2g$. What is its final velocity in terms of g ?



g downward

RECOGNITION CUE

MODEL / EQUATIONS

FIRST ALGEBRA STEP

FINAL VELOCITY

HINT LADDER

H1: free fall from rest.
 H2: $\bar{v}=gt/2$.
 H3: set $gt/2=g \rightarrow t=2\text{ s}$.

HINT LADDER

Use the same $\bar{v}=gt/2$ expression; solve time first; then $v=gt$.

M-ST11A

Independent transfer: average velocity as a number

Build the relation from free-fall physics instead of recalling a shortcut



g downward

QUESTION

A free-fall problem states that the average velocity has numerical value 14.7 in SI units while $g=9.8$. Find the elapsed time and final speed.

FIRST MOVE

Write $v=gt$ and $\bar{v}=gt/2$ first. Keep the numerical comparison separate from dimensional equality.

WORK ZONE

PHYSICAL RELATIONS

NUMERICAL CONDITION + TIME

FINAL SPEED + UNIT CHECK

M-ST11B

When velocity and displacement are numerically equal

Same number does not mean same physical quantity

$$20 \neq 20$$

m/s m

same number • different units

$$v = gt \quad \text{and} \quad s = \frac{1}{2}gt^2$$

free fall from rest

DERIVATION

Numerical equality in SI: $gt = \frac{1}{2}gt^2$. Keep the $t=0$ root visible, then for the nonzero case $t=2$ s and $v=2g$.

FIRST MOVE

Write both physical expressions with units visible. Equate only their numerical values when the problem explicitly states that relation.

COMMON TRAP

"Numerically equal" does not make metres equal to metres per second. Also do not cancel t before noticing the $t=0$ solution.

MEMORY ANCHOR

Variant: translate ratios the same way. Example numerical $v = 2 \times$ numerical $s \rightarrow gt = 2(\frac{1}{2}gt^2)$, then solve the nonzero root.

M-ST11B

Translate the numerical relation, then choose the physical root

The algebra is simple only after the wording is modelled correctly

GUIDED 1 • FULL

In SI units, a freely falling body from rest has the same numerical values of speed and displacement. Find the time.

GUIDED 2 • FADED

In SI units, the numerical speed is twice the numerical displacement. Find the nonzero time.

MODEL → CALCULATE

$$v=gt, s=\frac{1}{2}gt^2.$$

Set numerical values equal: $gt=\frac{1}{2}gt^2$.

Solutions include $t=0$; the nonzero time is $t=2$ s.

RECOGNITION CUE

MODEL / EQUATIONS

FIRST ALGEBRA STEP

NONZERO ROOT

HINT LADDER

H1: "same numerical values".

H2: write gt and $\frac{1}{2}gt^2$.

H3: exclude $t=0$ only after identifying it.

HINT LADDER

Translate "twice" as $gt = 2(\frac{1}{2}gt^2)$, then solve the nonzero root.

M-ST11B

Independent transfer: numerical displacement is three times speed

Translate the ratio condition without confusing dimensions

QUESTION

A problem says, in SI units, that the numerical displacement is three times the numerical speed during free fall from rest. Form the equation and solve for nonzero time.

FIRST MOVE

Write the physical expressions first, translate the numerical ratio second, and keep the $t=0$ root visible before selecting the nonzero case.

WORK ZONE

$V(T)$ AND $S(T)$

NUMERICAL RATIO EQUATION

ROOTS + PHYSICAL SELECTION

M-ST11C

Air resistance proportional to mass: why mass cancels

Foundation extension: cancel only after writing the stated force model



$$ma = mg - km$$

downward positive

$$a = g - k$$

the common factor m cancels

PHYSICAL CHECK

If m doubles, both mg and km double. Net force doubles too, so acceleration remains unchanged.

FIRST MOVE

Draw the force arrows and write Newton's second law before cancelling anything.

MODEL GATE

Mass cancels only for the stated model $F_{\text{res}} = k m$ (or a fixed fraction of mg). If resistance is a constant force R , then $a = g - R/m$ and mass does not cancel.

MEMORY ANCHOR

If every force term carries the same factor m , divide the whole equation by m . Do not generalise this to arbitrary drag laws.

M-ST11C

Show the force model before the cancellation

Guided examples separate a valid common factor from a memorised claim

GUIDED 1 • FULL

A body falls with downward weight mg and upward resistance $0.25mg$. Find its acceleration.

GUIDED 2 • FADED

Two masses m and $3m$ fall under gravity with upward resistance km . Compare their accelerations.



FORCE MODEL

COMMON FACTOR

ACCELERATION COMPARISON

MODEL → CALCULATE

Net downward force = $mg - 0.25mg = 0.75mg$.
 $ma = 0.75mg \rightarrow a = 0.75g$.

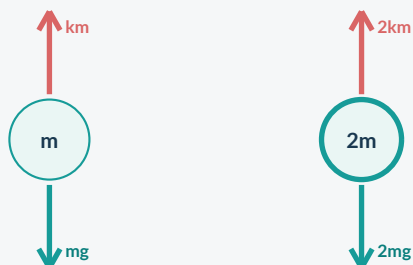
HINT LADDER

Write $ma = m(g-k)$ for each body. Compare only after the common mass factor is removed.

M-ST11C

Independent transfer: explain mass cancellation

State the force-model assumption before generalising the result



QUESTION

A model gives downward weight mg and upward resistance cm , where c has acceleration units. Explain why two different masses have the same acceleration, then state one drag model where they would not.

FIRST MOVE

Draw forces, write $ma = \Sigma F$, then inspect the common factor. The conclusion is conditional on the stated resistance law.

WORK ZONE

FORCE DIAGRAM / EQUATION

WHY M CANCELS

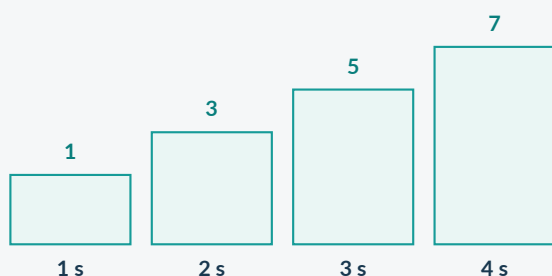
COUNTEREXAMPLE DRAG MODEL

M-ST11D

Ratio of nth-second distance to total distance

Build the ratio from interval and cumulative quantities; do not memorise another formula

successive interval factors: 1, 3, 5, 7



$$S_n = \frac{1}{2}an^2$$

total through n seconds

$$s_n = \frac{1}{2}a(2n-1)$$

distance during the nth second

$$s_n / S_n = (2n-1)/n^2$$

form the ratio only after naming both quantities

FIRST MOVE

Write cumulative S and interval s separately. Only then form the requested ratio.

BOUNDARY

The pure odd-number form assumes motion from rest with constant acceleration and equal time intervals. If $u \neq 0$, rebuild from the general interval difference.

MEMORY ANCHOR

Check at $n=4$: interval factors 1,3,5,7 sum to 16, so $s_4/S_4=7/16$. Consecutive interval ratio: $s_{-1}/s=(2n-3)/(2n-1)$.

M-ST11D

Compare intervals with totals without losing the boundary

Guided 1 uses the ratio directly; Guided 2 compares two interval distances

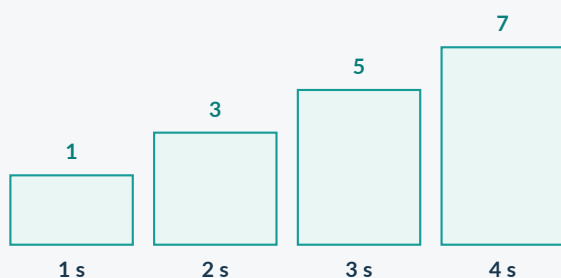
GUIDED 1 • FULL

From rest under constant acceleration, what fraction of the first 5-second total displacement occurs during the 5th second?

GUIDED 2 • FADED

For motion from rest, find s_7/s_6 without first finding a .

successive interval factors: 1, 3, 5, 7



INTERVAL MODEL

COMMON FACTOR

RATIO

MODEL → CHECK

s_5 has factor $2(5)-1=9$. s_5 has factor $5^2=25$. Fraction = $9/25$.

HINT LADDER

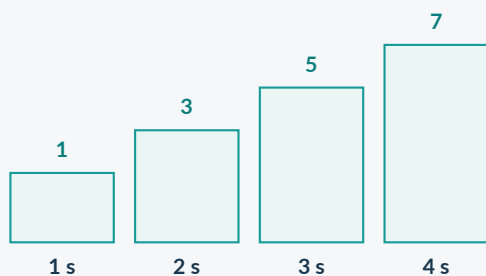
Both are interval distances: $s = \frac{1}{2}at^2$. Use $13/11$.

M-ST11D

Independent transfer: when does an interval exceed one-fifth?

Compare n th-second interval displacement with the cumulative total

successive interval factors: 1, 3, 5, 7



QUESTION

A body starts from rest with constant acceleration. During which second does the interval displacement first exceed one-fifth of the total displacement accumulated up to that second?

FIRST MOVE

Build $s/S = (2n-1)/n^2$ from the two quantities, then solve the inequality. Do not use the from-rest shortcut outside its model gate.

WORK ZONE

RATIO MODEL

INEQUALITY

FIRST INTEGER N + CHECK

M-ST12A

Rebuild the equation family from two ideas

Consolidation: one constant-acceleration model, four connected faces

ONE CONSTANT-ACCELERATION MODEL

$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

$$s = (u+v)t/2$$

$$v^2 = u^2 + 2as$$

IDEA 1 • ACCELERATION

$a=(v-u)/t \rightarrow v=u+at$. Physical story: final velocity = starting velocity + accumulated change.

IDEA 2 • V-T AREA

Displacement is area under v-t. Rectangle ut + triangle $\frac{1}{2}(at)t$ makes both the $\frac{1}{2}$ and t^2 visible.

MODEL GATE

These relations belong to one-dimensional motion with constant acceleration. If acceleration is not constant, this family is not automatically valid.

MEMORY ANCHOR

Do not memorise four unrelated formulas. Remember acceleration definition + v-t area, then rebuild the face you need.

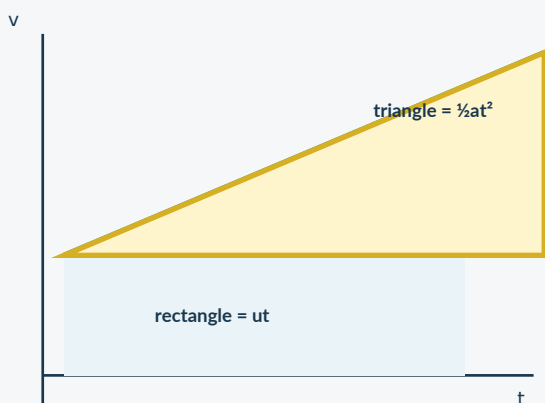
M-ST12A

Rebuild instead of recall

Use representation and elimination to derive two faces of the same model

GUIDED 1 • FULL SUPPORT

A car starts at u and accelerates uniformly at a for time t . Rebuild its displacement relation from a v - t graph.



DERIVATION

1. Rectangle area = ut .
2. Acceleration adds velocity at , so triangle height = at .
3. Triangle area = $\frac{1}{2} \times t \times at = \frac{1}{2}at^2$.
4. Total displacement $s = ut + \frac{1}{2}at^2$.

GUIDED 2 • FADED

Starting from $v=u+at$ and $s=(u+v)t/2$, eliminate t to recover the time-free relation.

SUBSTITUTE $t =$

SIMPLIFY $(U+V)(V-U)$

FINAL RELATION

M-ST12A

Independent transfer: rebuild all four faces

Use two starting ideas, then name the variable each relation avoids

QUESTION

Without looking back, reconstruct all four constant-acceleration relations from the two starting ideas. Next to each relation, write what variable it avoids.

FIRST MOVE

If one step is forgotten, return to meaning: acceleration is velocity change per time; displacement is area under v - t .

WORK ZONE

STARTING IDEA 1

STARTING IDEA 2

FOUR REBUILT RELATIONS + VARIABLES AVOIDED

CONSTANT-A MODEL GATE

M-ST12B

Which equation should I use?

Model first; then choose the relation that leaves one unnecessary variable absent

IF THIS IS ABSENT	USE	WHY
s absent	$v = u + at$	no displacement needed
v absent	$s = ut + \frac{1}{2}at^2$	no final velocity needed
t absent	$v^2 = u^2 + 2as$	direct speed-distance
a absent	$s = (u+v)t/2$	endpoint velocities + time
nth second	$S(n)-S(n-1)$	interval, not cumulative

DECISION ROUTINE

WORDS → MODEL → SIGNS → TRANSLATE → TARGET → EXCLUDE → CHOOSE → CHECK. Rest means $u=0$; stop means $v=0$. Circle the target before choosing the relation.

CORE RULE

The matrix is an aid after modelling, not a substitute for modelling. Constant acceleration must be established before the standard equation family is used.

MEMORY ANCHOR

Choose the relation that keeps the unnecessary variable absent. If the problem is an interval, graph, delayed-start or two-body structure, classify that first.

M-ST12B

Make formula choice visible

Cross out what you do not need, then check the special-structure anti-triggers

EXAMPLE A • STOPPING TIME

Car: $u=20 \text{ m/s}$, $a=-5 \text{ m/s}^2$, $v=0$. Target t ; displacement s is unnecessary.

Choose $v=u+at$.

EXAMPLE B • STOPPING DISTANCE

Same car and braking. Target s ; time t is unnecessary.

Choose $v^2=u^2+2as$.

ANTI-TRIGGERS

NONCONSTANT a : standard constant- a relations are not automatic.

TWO-BODY / DELAYED START: build the common timeline or relative model first.

n th SECOND: use $S(n)-S(n-1)$, not cumulative $S(n)$.

GRAPH QUESTION: read slope/area directly when the graph already carries the physics.

GUIDED 1

A train at 12 m/s accelerates uniformly at 2 m/s^2 for 5 s . Find displacement.

GUIDED 2

A body speeds from 8 to 20 m/s over 84 m with constant acceleration. Find a .

ROUTE

Constant a ; v is not needed $\rightarrow$ use $s=ut+\frac{1}{2}at^2$.

KNOWN

UNNECESSARY VARIABLE

EQUATION

M-ST12B

Independent mixed transfer: choose the route

Do not solve; justify model → representation → equation

QUESTION

For each mini-story, do not solve. Only name the model, target, unnecessary variable, and first equation: (a) braking distance, (b) delayed second car meets first, (c) 7th-second distance, (d) v-t graph displacement.

FIRST MOVE

When stuck: MODEL → SIGNS → TRANSLATE → TARGET → EXCLUDE → CHOOSE → CHECK. Special structures come before equation selection.

WORK ZONE

A • BRAKING DISTANCE

B • DELAYED-START MEETING

C • 7TH-SECOND DISTANCE

D • V-T GRAPH DISPLACEMENT

M-ST12C

Reconstruction test: can you explain the physics?

Final consolidation: explain the shape of the equation before calculating

 t^2

Why does acceleration contribute t^2 to displacement?

 v^2

Why does the time-free relation contain squared velocities?

 u^2

Why is stopping distance proportional to u^2 ?

1,3,5

Why do equal-time distances from rest follow odd numbers?

 $v=0$

How can $v=0$ while a is not zero?

 g cancels

Why can common gravity disappear from relative motion?

area

Why does v - t area have displacement units?

RULE

Answer in ordinary language first. Then rebuild the equation or graph consequence.

WHY THIS TEST

It reveals whether you own the structure or only remember formula labels.

MEMORY ANCHOR

If you can explain why an equation or graph has that shape, you can rebuild it when memory fails.

M-ST12C

Seven ideas, three reconstruction stories

Connect mathematical fingerprints to physical mechanisms

WHY T^2 ?

Acceleration makes velocity grow with time. Displacement accumulates that growing velocity, so the acceleration-added part grows like time $\times$ time. The v - t triangle gives $\frac{1}{2}at^2$.

WHY V^2 ?

Use $t=(v-u)/a$ and $s=(u+v)t/2$. Eliminating t multiplies $(u+v)(v-u)$, producing v^2-u^2 : the algebraic fingerprint of removing time.

STOPPING + ODD DISTANCES

With $v=0$, $v^2=u^2+2as$ gives $s_{\text{stop}}u^2$ for fixed braking. From rest, cumulative t^2 totals have first differences 1,3,5,...

TURNING POINT + RELATIVE G

At the top, $v=0$ but $a=-g$ remains, causing reversal. For two bodies with the same gravity, $a_{\text{rel}}=(-g)-(-g)=0$ even though gravity still acts on each.

WHY V-T AREA?

Velocity $\times$ time has displacement units: $(\text{m/s})\times\text{s}=\text{m}$. Signed areas above and below zero contribute opposite displacement signs; distance uses absolute area magnitudes.

M-ST12C

Guided reconstruction and final teach-back

No formula labels as a substitute for explanation

GUIDED 1

Why does doubling initial speed quadruple ideal stopping distance under the same braking acceleration?

GUIDED 2 • FADED

Explain why successive one-second distances from rest form an arithmetic sequence without quoting the n th-second formula.

HINT LADDER

H1: stop $\rightarrow v=0$.

H2: time is unnecessary.

H3: solve $v^2 = u^2 + 2as$ for s and inspect the power of u .

CUMULATIVE LAW

FIRST DIFFERENCES

CONCLUSION

FINAL TRANSFER • TEACH IT BACK

Choose any three reconstruction prompts. Explain each to a younger student using a physical story, a diagram or graph, and at most one equation.

PASS STANDARD

Each explanation must include: (1) the model assumption, (2) the physical mechanism, and (3) one check or counterexample.

PROMPT 1 • EXPLAIN / DRAW

PROMPT 2 • EXPLAIN / DRAW

PROMPT 3 • EXPLAIN / DRAW
